

Convex relaxation gaps and spatial certificates in nonlinear optimization

Abstract

Separate convexification of nonlinear terms can lose information about their shared variables. We develop a probability formulation of this loss and explain its relation to spatial optimality certificates. For bilinear functions with arbitrary coefficient signs, an elementary cut argument bounds the ratio of the McCormick gap to the scalar graph-hull gap by four times the square root of the largest induced edge density. A random-sign construction gives the matching order on each individual interaction graph. Sharper maximum-degree and bipartite bounds, exactness criteria, and the distinction between exact termwise convexification and recursive product relaxations identify the scope of these comparisons. The density order is also a consequence of earlier Schur-multiplier theory; the explicit transfer and proof are included to make that attribution precise.

1 Introduction

Convex relaxations support global optimization by providing bounds on nonlinear functions over a region. Their strength depends on how the model is divided into pieces. Even when each piece is replaced by its exact convex hull, the resulting bounds can be weaker than the hull of their sum: separate pieces may use incompatible representations of their common variables. Spatial subdivision introduces a second question, namely how many regions are needed before a specified bounding rule certifies a desired tolerance. These questions concern different mathematical objects. A pointwise envelope gap comparison is not an approximation ratio for a constrained optimization problem, and it does not by itself bound a branch-and-bound tree.

McCormick’s construction provides the basic recursive mechanism for factorable functions [5]. For multilinear functions, the convex hull of the scalar graph has a finite representation using box vertices. Luedtke, Namazifar, and Linderoth [4] compare this hull with termwise and recursive relaxations, and establish several exactness and positive-coefficient bounds. Boland et al. [1] show that signed bilinear gaps can grow with the square root of dimension and relate the comparison to cuts in induced interaction graphs.

We use the vertex representation to state the comparisons consistently. For positive multilinear functions on nonnegative boxes, a common probability distribution attains all upper term envelopes, so the entire gap question reduces to a single simultaneous rounding problem. For signed bilinear functions, an exact cut-range characterization instead permits graph density and degree bounds. The elementary density proof makes the constants explicit, while the accompanying Schur-multiplier transfer explains which asymptotic content was already implicit in established theory [2]. The framework also specifies the bounding oracle and the certified-region count needed for spatial lower bounds.

2 Envelope gaps as probability problems

2.1 The scalar graph and the specified termwise relaxation

Let $B = \prod_{i=1}^n [\ell_i, u_i]$ be a finite box. Fixed coordinates are removed when necessary. A function is *multiaffine* if it is affine in each coordinate separately; equivalently, it is a multilinear polynomial with no repeated variable in a monomial. For a continuous scalar function f on B , let $\text{vex}_B f$ be its greatest convex underestimator and $\text{cav}_B f$ its least concave overestimator. The vertical section of the convex hull of its graph is

$$\{y : (x, y) \in \text{conv}\{(z, f(z)) : z \in B\}\} = [\text{vex}_B f(x), \text{cav}_B f(x)].$$

Define the *scalar graph-hull gap* and, for a specified decomposition $f = \sum_{e \in E} f_e$, the *termwise gap* by

$$H_B(f; x) = \text{cav}_B f(x) - \text{vex}_B f(x), \quad (2.1)$$

$$T_B(f; x) = \sum_{e \in E} [\text{cav}_B f_e(x) - \text{vex}_B f_e(x)]. \quad (2.2)$$

The notation T_B always refers to the displayed decomposition. It is the width obtained by giving each scalar factor its exact graph hull and imposing $y = \sum_e y_e$. It can change when a factor is split into smaller pieces, whereas H_B does not. Always,

$$0 \leq H_B(f; x) \leq T_B(f; x), \quad (2.3)$$

because the sum of convex underestimators is a convex underestimator of the sum, and similarly for concave overestimators. Ratios are taken only where $H_B(f; x) > 0$; gap inequalities apply at every point.

For a positive multilinear polynomial, the factors are its displayed monomials $a_e \prod_{i \in e} x_i$, with $a_e \geq 0$. For a signed bilinear function, they are the signed products $a_{ij} x_i x_j$. A bilinear factor has the familiar four-inequality McCormick hull. For a higher-degree factor, its exact hull and a recursive McCormick relaxation are different objects. A recursive product relaxation is exact for one monomial on a unit box, but need not be exact on a general positive box [4, Sections 2–3]. A bound for (2.2) on that box therefore does not automatically bound the gap of a recursive bilinear factorization.

Adding an affine function to a factor, or adding a separate affine factor, changes neither gap. An affine bijection between boxes preserves both envelopes after composition and hence preserves the gaps for the transformed factors. These facts follow from the graph-hull representation.

2.2 Vertex distributions and compatible marginals

Let $\mathcal{V}(B)$ denote the vertices of B , and let $\mathcal{P}_B(x)$ be the distributions on $\mathcal{V}(B)$ with mean x . On the unit cube these are joint Bernoulli distributions with the prescribed singleton means; coordinates need not be independent.

Lemma 2.1 (Vertex representation). *For a multiaffine f on B ,*

$$\text{vex}_B f(x) = \min_{P \in \mathcal{P}_B(x)} \mathbb{E}_P f(X), \quad (2.4)$$

$$\text{cav}_B f(x) = \max_{P \in \mathcal{P}_B(x)} \mathbb{E}_P f(X). \quad (2.5)$$

Both extrema are attained. The envelopes are continuous piecewise-affine functions on B .

Proof. Write $x_i = \ell_i + (u_i - \ell_i)p_i$. Independently choose $X_i = u_i$ with probability p_i and $X_i = \ell_i$ otherwise. Multiaffinity gives $\mathbb{E}X = x$ and $\mathbb{E}f(X) = f(x)$. Every graph point therefore belongs to the convex hull of vertex graph points, and the reverse inclusion in the full graph hull is immediate. The vertical sections give the formulas. The feasible distributions form a nonempty compact polytope, so both extrema are attained. The lower and upper surfaces of the vertex graph polytope are respectively maxima and minima of finitely many affine functions on B , which also proves continuity at its boundary. $\square$

These representations are classical [4, Section 3]. They expose the compatibility issue: different factors may use different minimizing distributions in (2.2), whereas (2.4) requires one distribution for their sum.

For $m_e(x) = \prod_{i \in e} x_i$ with $e \neq \emptyset$, the unit-cube formulas are

$$\text{cav } m_e(x) = \min_{i \in e} x_i, \quad \text{vex } m_e(x) = \max\{0, \sum_{i \in e} x_i - |e| + 1\}. \quad (2.6)$$

To prove them, interpret $\mathbb{E}m_e(X)$ as the probability of joint success. The upper bound is attained by $X_i = \mathbf{1}\{U \leq x_i\}$ for one uniform $U \in [0, 1]$. For the lower bound, place failure intervals of lengths $1 - x_i$ consecutively on the unit circle. If their total length is below one they are disjoint, and if it is at least one they cover the circle. Their union therefore has measure $\min\{1, \sum_{i \in e} (1 - x_i)\}$. Each interval has its prescribed measure, so the complement gives the claimed minimum joint-success probability. Unused coordinates can be sampled independently to complete the mean vector.

Proposition 2.2 (Common upper attainment and deficiencies). *Let $f(x) = \sum_e a_e m_e(x)$ on $[0, 1]^n$, with $a_e \geq 0$. Discard supports of size at most one. For each remaining support choose an anchor i_e of smallest mean and put*

$$u_e = x_{i_e}, \quad S_e = \sum_{j \in e \setminus \{i_e\}} (1 - x_j), \quad t_e = \min\{u_e, S_e\}, \quad D_e(X) = X_{i_e} - \prod_{i \in e} X_i.$$

Then D_e is the nonnegative indicator that the anchor succeeds and some other support coordinate fails. Moreover,

$$\text{cav } f(x) = \sum_e a_e u_e, \quad (2.7)$$

$$T(f; x) = \sum_e a_e t_e, \quad (2.8)$$

$$H(f; x) = \max_{P \in \mathcal{P}_{[0,1]^n}(x)} \sum_e a_e \mathbb{E}_P D_e(X). \quad (2.9)$$

Proof. The common-threshold law attains every single-product upper envelope simultaneously. Positivity gives (2.7), and subtracting the two single-product envelopes gives (2.8). Finally, every admissible law has $\mathbb{E}X_{i_e} = u_e$, so minimizing the expected polynomial value is equivalent to maximizing the weighted sum of deficiencies. $\square$

The common-upper statement is classical [4, Theorems 4–5 in the authors’ manuscript]. A distribution used to prove a gap bound must work for all terms together; independently optimizing the distribution for each term merely recovers the termwise relaxation. Mixtures of admissible laws preserve the means, and nonnegative deficiencies allow the guarantees of different laws to be added.

Lemma 2.3 (Two classes handled by independence). *Put $c_0 = 1 - e^{-1}$. Independent Bernoulli rounding gives $\mathbb{E}D_e \geq c_0 t_e/2$ if either every support mean exceeds $1/2$, or at least two support means are at most $1/2$.*

Proof. Independence gives $\mathbb{E}D_e = u_e(1 - \prod_{j \neq i_e} x_j)$. In the second case a nonanchor mean is at most $1/2$, giving at least $u_e/2 \geq t_e/2$. In the first case the expression is at least $u_e(1 - e^{-S_e}) \geq c_0 u_e \min(1, S_e)$. If $S_e \leq 1$, this is at least $c_0 S_e/2 \geq c_0 t_e/2$; otherwise it is at least $c_0 u_e = c_0 t_e$. $\square$

Proposition 2.4 (Nonnegative-box transfer). *For a positive multilinear polynomial on B with $0 \leq \ell_i < u_i$, substitute $x_i = \ell_i + (u_i - \ell_i)p_i$ and expand the original monomials. The unit-cube polynomial f has nonnegative coefficients and no larger degree. With its expanded monomials as factors,*

$$H_B(f; x) = H(\tilde{f}; p), \quad T_B(f; x) \leq T(\tilde{f}; p). \quad (2.10)$$

One common-threshold endpoint law attains the upper envelope of every original monomial and their sum.

Proof. Expansion coefficients are products of nonnegative endpoints and interval widths. The full-envelope equality follows from the affine bijection. For an original factor expanded as $\sum_q g_q$, the sum of the lower convex envelopes is a convex underestimator and the sum of the upper concave envelopes is a concave overestimator. Subtract and sum over the original factors to obtain (2.10). Common-threshold rounding attains all expanded upper envelopes together and therefore also those of each original factor and their sum. $\square$

Expansion is a proof device and need not be an efficient representation. It changes the factor incidence structure. Degree bounds transfer through (2.10); bounds based on the original incidence graph require separate arguments. For bilinear functions, arbitrary box endpoints cause no such difficulty: an affine change introduces only scaled bilinear terms and affine summands, regardless of signs.

2.3 What a spatial certificate counts

Let $\mathcal{F}$ be a nonempty compact feasible set, let f be continuous, and put $f^* = \min_{x \in \mathcal{F}} f(x)$. A region oracle assigns a valid lower bound $\text{LB}(S)$ to feasible points in each region S . A *certificate at target* T_* is a finite cover $S_1, \dots, S_N$ of $\mathcal{F}$ with $\text{LB}(S_j) \geq T_*$ for all j . Together with a feasible value U and an absolute tolerance $\epsilon \geq 0$, the target $U - \epsilon$ certifies absolute gap at most ϵ . For $U > 0$ and a relative tolerance $0 \leq \theta < 1$, the target $(1 - \theta)U$ certifies $(U - f^*)/U \leq \theta$. A size lower bound may grant the best incumbent $U = f^*$, requiring $f^* > 0$ in this relative convention. Under these tolerance ranges, either target is nondecreasing in U , so worse incumbents require weakly higher targets.

A complete spatial split tree supplies a cover by its final regions. A lower bound for arbitrary covers is thus independent of branching order and node selection within the specified region and oracle model. If a procedure discards feasible regions by propagation or additional cuts, those deductions must also be included in the certificate model. A lower bound for one oracle does not automatically survive stronger coupled cuts, different branch coordinates, or analytic solution of separate components.

3 Signed bilinear gaps and interaction density

Let $G = (V, E)$ be a finite simple graph and $b_a(x) = \sum_{ij \in E} a_{ij} x_i x_j$ on $[0, 1]^V$, with arbitrary real coefficients. Zero-coefficient edges may be removed. Let $c^*(a)$ be the least $c \geq 0$ with $T(b_a; x) \leq cH(b_a; x)$ everywhere, and set $c^*(0) = 0$. Define

$$\rho_G(G) = \max_{\emptyset \neq W \subseteq V} \frac{|E(G[W])|}{|W|}, \quad (3.1)$$

with value zero for an empty graph. This is half the maximum subgraph average degree, not fractional arboricity, whose denominator is $|W| - 1$. Let d_G be degeneracy and Δ_G maximum degree. Then

$$d_G/2 \leq \rho_G(G) \leq d_G, \quad \rho_G(G) \leq \Delta_G/2. \quad (3.2)$$

The first lower bound uses a subgraph of minimum degree d_G ; the upper bounds use a degeneracy orientation and the degree sum identity.

Theorem 3.1 (Bilinear upper bounds). *For every real coefficient vector on G ,*

$$c^*(a) \leq \min\{4\sqrt{\rho_G(G)}, 4\sqrt{d_G}, 2\sqrt{\Delta_G}\}. \quad (3.3)$$

If G is bipartite with parts P, Q and has an edge, let Δ_P and Δ_Q be the maximum degrees in the respective parts. Then

$$c^*(a) \leq \sqrt{2 \min\{\Delta_P, \Delta_Q\}} \leq \sqrt{2 \min\{|P|, |Q|\}}. \quad (3.4)$$

The factor $\sqrt{2}$ in the first bipartite bound is sharp. The conclusions transfer to arbitrary finite boxes after removing fixed coordinates.

The proof refines the cut argument of Boland et al. [1]. The density order also follows from older operator theory, as shown in Section 3.6; no new general graph-norm principle is claimed.

3.1 An exact characterization by induced cuts

For $W \subseteq V$, set

$$Q_{a,W}(s) = \sum_{ij \in E(G[W])} a_{ij} s_i s_j, \quad L_W(a) = \sum_{ij \in E(G[W])} |a_{ij}|,$$

and define

$$R_W(a) = \frac{1}{2} \left(\max_{s \in \{-1,1\}^W} Q_{a,W}(s) - \min_{s \in \{-1,1\}^W} Q_{a,W}(s) \right). \quad (3.5)$$

A cut encoded by s has weight $\frac{1}{2} \sum_{ij} a_{ij} - \frac{1}{2} Q_{a,W}(s)$, so R_W is exactly the maximum cut weight minus the minimum cut weight. If $L_W > 0$ then $R_W > 0$: distinct degree-two sign characters are orthogonal under uniform signs, and their nonzero combination cannot be constant. Set $L_W/R_W = 0$ when both vanish.

Lemma 3.2 (Induced-cut characterization). *For every coefficient vector,*

$$c^*(a) = \max_{W \subseteq V} \frac{L_W(a)}{R_W(a)}. \quad (3.6)$$

Each nonzero induced ratio is attained at $x_i = 1/2$ on W and $x_i = 0$ outside W .

Proof. At that point any admissible vertex law fixes outside coordinates to zero. Write $X_i = (1+s_i)/2$ on W . Since $\mathbb{E}s_i = 0$,

$$\mathbb{E}b_a(X) = \frac{1}{4} \left(\sum_{ij \in E(G[W])} a_{ij} + \mathbb{E}Q_{a,W}(s) \right).$$

An extremizing sign vector and its negative, each of probability $1/2$, attain either extreme of $Q_{a,W}$ with the prescribed means. Thus $H = R_W/2$, whereas each active product has width $1/2$ and

$T = L_W/2$. These identities hold at every point of $\{0, 1/2, 1\}^V$, with W its set of half-valued coordinates: fixing coordinates to one instead of zero adds only affine terms on W .

To extend the comparison, the width of the product $x_i x_j$ is

$$\min(x_i, x_j) - \max(0, x_i + x_j - 1) = \min(x_i, x_j, 1 - x_i, 1 - x_j).$$

Partition the cube by the hyperplanes $x_i = x_j$, $x_i + x_j = 1$ ($i \neq j$), and $x_i = 1/2$. The termwise gap is affine on every closed cell. Every cell vertex is in $\{0, 1/2, 1\}^V$. Indeed, active equations fix a coordinate to 0, 1/2, or 1, or relate two coordinates by equality or complementation. In a connected component of these relations, one fixed coordinate determines all values. If none is fixed, a cycle with an odd number of complementations forces the value 1/2. Otherwise a free parameter remains and a small perturbation preserves all active equations, contradicting extremality. This accounts for every cell vertex.

The graph-hull gap is concave, being $\text{cav } b_a - \text{vex } b_a$. Hence $T - cH$ is convex on each cell for $c \geq 0$. Its value at any convex combination of cell vertices is at most the combination of its values there. The grid-point bound therefore proves the upper direction of (3.6), and the attained ratios prove the other. $\square$

The grid identities and upper implication are those of Boland et al. [1, Lemma 1 and Corollary 1], following Luedtke et al. [4]. The cell argument above supplies the boundary details.

3.2 Cut ranges, row norms, and fractional orientations

Let A be the symmetric weighted adjacency matrix on W , with zero diagonal and zero entries at nonedges. Write $a_{i,T} = (A_{ij})_{j \in T}$ for $T \subseteq W$ and $a_i = a_{i,W}$. For a rectangular matrix define $\|M\|_{\infty \rightarrow 1} = \max_u \text{sign} \|Mu\|_1 = \max_{u,v} \text{sign } v^\top Mu$; empty matrices have norm zero.

Lemma 3.3 (Polarization and row estimates). *Writing $R = R_W(a)$,*

$$R = \max_{S \subseteq W} \|A_{S,W \setminus S}\|_{\infty \rightarrow 1}, \quad (3.7)$$

$$R \geq \frac{1}{\sqrt{2}} \max_{S \subseteq W} \sum_{i \in S} \|a_{i,W \setminus S}\|_2, \quad (3.8)$$

$$R \geq \frac{1}{4} \sum_{i \in W} \|a_i\|_2. \quad (3.9)$$

Proof. For signs s, t , put $u = (s + t)/2$, $v = (s - t)/2$. Their supports are complementary, and symmetry gives $Q(s) - Q(t) = 2v^\top Au$. Conversely, arbitrary signs on complementary supports arise from $s = u + v$, $t = u - v$. Maximizing half the difference proves (3.7).

The sharp real Khinchin inequality of Szarek [7] states that independent uniform signs satisfy

$$\mathbb{E} \left| \sum_j c_j \varepsilon_j \right| \geq \frac{1}{\sqrt{2}} \left(\sum_j c_j^2 \right)^{1/2}. \quad (3.10)$$

Apply it to every row of $A_{S,T}$ for a fixed partition S, T , and sum. The best sign vector performs at least as well as its average, proving (3.8). Transposition gives the same estimate for side T .

Choose a partition locally maximizing the cut weight with nonnegative weights a_{ij}^2 . At every vertex at least half its squared incident weight crosses, since otherwise moving the vertex improves the cut. Thus

$$X = \sum_{i \in S} \|a_{i,T}\|_2, \quad Y = \sum_{i \in T} \|a_{i,S}\|_2 \quad \text{satisfy} \quad X + Y \geq \frac{1}{\sqrt{2}} \sum_i \|a_i\|_2.$$

The two row estimates yield $R \geq \max(X, Y)/\sqrt{2} \geq (X + Y)/(2\sqrt{2})$, which proves (3.9). $\square$

Lemma 3.4 (Fractional orientation). *Every finite graph H admits nonnegative fractions $\theta_{ij} + \theta_{ji} = 1$ on each edge such that $\sum_j \theta_{ij} \leq \rho_G(H)$ for every vertex. This is the least possible uniform bound on the vertex loads.*

Proof. For $m = |E(H)| > 0$, form a flow network with a source, edge nodes, vertex nodes, and a sink. Source-to-edge capacities are one; arcs from an edge node to its endpoints have capacity $m + 1$; vertex-to-sink capacities are t . A cut of capacity at most m cannot cut an edge-to-endpoint arc. For a fixed set U of vertex nodes on the source side, placing all edge nodes internal to U on that side minimizes capacity, giving $m - |E(H[U])| + t|U|$. At $t = \rho_G(H)$ every cut has capacity at least m . Max-flow/min-cut supplies a flow saturating the source arcs; outgoing flows at each edge define its fractions. Conversely, the loads over U sum to at least $|E(H[U])|$, proving minimality. For $m = 0$ use zero loads. $\square$

Proof of Theorem 3.1. Fix an induced graph H with nonempty weighted support and abbreviate $L = \sum_{ij} |a_{ij}|$, $R = R_{V(H)}(a)$. Weighted Cauchy–Schwarz gives

$$\sum_j \theta_{ij} |a_{ij}| \leq \left(\sum_j \theta_{ij} \right)^{1/2} \left(\sum_j \theta_{ij} a_{ij}^2 \right)^{1/2} \leq \sqrt{\rho_G(H)} \|a_i\|_2.$$

Summing counts each absolute edge weight once. By (3.9), $L \leq 4\sqrt{\rho_G(H)}R$. Also $\|a_i\|_2 \geq \|a_i\|_1/\sqrt{\Delta_H}$ at every nonisolated vertex, and $\sum_i \|a_i\|_1 = 2L$. The same row bound therefore gives $R \geq L/(2\sqrt{\Delta_H})$. The graph parameters do not increase in induced subgraphs, so Lemma 3.2 proves (3.3).

For bipartite H use its inherited parts P', Q' in (3.8). Every edge crosses, giving

$$R \geq \frac{1}{\sqrt{2}} \sum_{i \in P'} \|a_i\|_2 \geq \frac{L}{\sqrt{2\Delta_P}}.$$

Transpose to obtain the bound with Δ_Q and apply the same induced characterization. On a unit-weight four-cycle with one negative edge, $L = 4$, $R = 2$, and $\Delta_P = \Delta_Q = 2$. Its center ratio attains two, so the stated bipartite constant is sharp. An affine box transformation scales each bilinear coefficient by the two interval widths and introduces only affine summands, proving the final assertion. $\square$

The degeneracy bound has a distinct direct proof. Orient along a degeneracy order with outdegree at most d_G and color vertices in reverse order. Place a vertex opposite the side receiving at least half the squared weight of its outgoing edges. Its crossing row norm is at least its outgoing row norm divided by $\sqrt{2}$. Summing and applying Cauchy–Schwarz to at most d_G outgoing entries gives $X + Y \geq L/\sqrt{2d_G}$. The final step of Lemma 3.3 yields $R \geq L/(4\sqrt{d_G})$. This proof is weaker than the density theorem but uses an ordinary acyclic orientation.

3.3 Sharp order on every interaction graph

Define

$$\Gamma(G) = \sup_{a \in \mathbb{R}^E \setminus \{0\}} c^*(a), \quad \Gamma_{\pm}(G) = \max_{a \in \{-1, 1\}^E} c^*(a),$$

with both values zero for edgeless graphs.

Theorem 3.5 (Worst coefficients on a fixed graph). *For every graph with an edge,*

$$\max \left\{ 1, \sqrt{\frac{\rho_G(G)}{2 \log 2}} \right\} \leq \Gamma_{\pm}(G) \leq \Gamma(G) \leq 4\sqrt{\rho_G(G)}. \quad (3.11)$$

Requiring every coefficient to be nonzero does not change $\Gamma(G)$. It also equals

$$\sup_{a \neq 0} L_V(a)/R_V(a). \quad (3.12)$$

The lower bound has a full-support signing and a witness in $\{0, 1/2\}^V$.

Proof. On an h -vertex, m -edge graph with $m > 0$, choose independent edge signs σ_e . For fixed vertex signs s and $\lambda > 0$, $\mathbb{E}e^{\lambda Q_{\sigma}(s)} = \cosh(\lambda)^m \leq e^{m\lambda^2/2}$, and similarly for $-\lambda$. Put $Z = \max_s |Q_{\sigma}(s)|$. Global sign reversal preserves Q_{σ} , so

$$e^{\lambda Z} \leq \sum_{s: s_1=1} (e^{\lambda Q_{\sigma}(s)} + e^{-\lambda Q_{\sigma}(s)}).$$

Taking expectations and using Jensen gives $\mathbb{E}Z \leq h \log 2/\lambda + m\lambda/2$. At $\lambda = \sqrt{2h \log 2/m}$, at least one signing satisfies

$$\|Q_{\sigma}\|_{\infty} \leq \sqrt{2mh \log 2}. \quad (3.13)$$

The values at different sign vectors need not be independent.

Apply this to a densest induced graph $H = G[W]$. Since $L_W = m$ and $R_W \leq \|Q_{\sigma}\|_{\infty}$, $L_W/R_W \geq \sqrt{\rho_G(G)/(2 \log 2)}$. Give all remaining edges arbitrary signs; they do not affect the witness face of Lemma 3.2. The lower bound one follows from $R_W \leq L_W$, and the upper bound is Theorem 3.1.

Call the supremum in (3.12) C . The full center shows $\Gamma \geq C$. Conversely, every induced ratio is a whole-graph ratio after zeroing all coefficients outside the induced subgraph. Thus $\Gamma \leq C$. On the compact set $L_V(a) = 1$, R_V is continuous and strictly positive, so the ratio attains a maximum. Perturb zero entries of a maximizer to nonzero values. Continuity of the full-graph ratio shows that the full-support supremum is also C . This does not assume continuity of c^* as edges disappear or attainment under the full-support restriction. $\square$

Uniform boundedness of all signed factors in a graph family is therefore equivalent to uniform boundedness of its induced density. No closure assumption on the family is needed: fixing outside coordinates exposes a densest subgraph. Global average density is insufficient. Attaching q^2 leaves to one vertex of $K_{q,q}$ produces a connected graph with total edge-to-vertex ratio less than two but induced density at least $q/2$.

The density is unweighted. Scaling all coefficients by a nonzero scalar preserves the gap ratio, whereas a sum of coefficient magnitudes scales. A single edge of arbitrarily small magnitude therefore rules out replacing ρ_G by $\max_W \sum_{e \in E(W)} |a_e|/|W|$. The row estimates remain coefficient-sensitive certificates; a uniform gap bound must use them on every induced subgraph, not only the full graph.

The order may equivalently be expressed using degeneracy or arboricity. If α_G is the minimum number of forests partitioning the edges, then $\rho_G(G) \leq \alpha_G \leq d_G \leq 2\rho_G(G)$. For the middle inequality, label outgoing edges in a degeneracy orientation with distinct labels from 1 to d_G . Each label class is a forest: an undirected cycle would have an earliest vertex with two outgoing edges of that label. The other inequalities follow by counting forest edges and (3.2).

3.4 Positive coefficients, exactness, and finite constants

For positive coefficients a proper coloring with $\chi \geq 2$ colors gives the classical stronger bound

$$c^*(a) \leq \begin{cases} 2 - 2/\chi, & \chi \text{ even,} \\ 2 - 2/(\chi + 1), & \chi \text{ odd.} \end{cases} \quad (3.14)$$

Choose uniformly $\lfloor \chi/2 \rfloor$ colors for one side of a cut. Every edge crosses with probability $p_\chi = 2\lfloor \chi/2 \rfloor \lceil \chi/2 \rceil / [\chi(\chi - 1)]$. Thus the maximum cut is at least $p_\chi L$, and the minimum cut is zero. The same coloring works on every induced subgraph, so Lemma 3.2 gives (3.14). This is the bound of Luedtke et al. [4, Theorem 8 in the authors' manuscript]. In particular, the positive bilinear factor is at most two and is one when its nonzero support is nonempty and bipartite. For the all-positive K_n with $n \geq 2$, the center ratio is $\binom{n}{2} / \lfloor n^2/4 \rfloor$, tending to two. Hence the worst-over-signs quantifier in Theorem 3.5 is essential.

Assume the effective support graph, obtained by deleting zero-coefficient edges, has at least one edge. Then $c^*(a) = 1$ if and only if every cycle of this graph has an even number of positive and an even number of negative edges [1, Theorem 4]. Boland et al. identify this as a consequence of the convex-envelope criterion in Theorem 3.10 of Misener, Smadbeck, and Floudas, *Dynamically generated cutting planes for mixed-integer quadratically constrained quadratic programs and their incorporation into GloMIQO 2*; our attribution to that predecessor is through Boland et al., who give the alternative cut proof used here. Indeed, $R_V = L_V$ exactly when one cut contains all positive edges and no negative edges, and another contains all negative edges and no positive edges. An edge set is a cut exactly when it intersects every cycle evenly. Necessity follows by traversing a cycle. For sufficiency, label vertices by the parity of specified edges on a path from a root in each component; cycle parity makes the label path-independent and supplies the cut. The criterion passes to induced subgraphs, proving sufficiency for $c^* = 1$. Conversely, $c^* = 1$ implies $R_V = L_V$ by Lemma 3.2. In particular, every forest with at least one edge has factor one for every signing. For empty effective support the gaps coincide at zero and $c^* = 0$ by convention.

The best universal constant in $\Gamma(G) \leq K_\rho \sqrt{\rho_G(G)}$ lies in $[2, 4]$: the lower bound is the four-cycle with one negative edge, with density one and factor two. For degeneracy the corresponding interval is $[\sqrt{2}, 4]$. These are finite-parameter questions, distinct from the best constants along sequences with density or degeneracy tending to infinity; the four-cycle does not give a growing-density lower bound.

The bounds imply $c^* \leq 4\sqrt{2}$ for series-parallel graphs by degeneracy, $c^* \leq 4\sqrt{3}$ for planar graphs by density, and $c^* \leq \sqrt{2 \min(p, q)}$ for $K_{p, q}$. For K_n , the maximum-degree bound is $2\sqrt{n - 1}$, compared with $600\sqrt{n}$ in Boland et al. [1, Theorem 2]. These are scalar vertical-gap comparisons, not approximation guarantees for a constrained objective.

3.5 Exact finite full-signing constants

For $n \geq 2$, distinguish the largest full-center ratio among complete-graph signings from the largest ratio over all points:

$$M_n = \max_{a \in \{-1, 1\}^{E(K_n)}} \frac{\binom{n}{2}}{R_V(a)}, \quad F_n = \Gamma_\pm(K_n).$$

All edges here have magnitude one. These maxima differ from the arbitrary-real-coefficient supremum in (3.12).

Proposition 3.6 (Finite exhaustive computation). *The exact values are*

n	3	4	5	6	7
$\min_a R_V(a)$	2	4	4	5	8
M_n	3/2	3/2	5/2	3	21/8
F_n	3/2	3/2	5/2	3	3

Proof. The first row is established by the exhaustive integer computation in Appendix A, which proves completeness of the enumeration, gives executable code, and supplies attaining signings. Dividing $\binom{n}{2}$ by that row gives M_n . Lemma 3.2 gives

$$F_n = \max_{2 \leq k \leq n} M_k.$$

Indeed, each induced support of a full signing is a full signing of some K_k , giving the upper bound. Conversely, any signing of K_k extends to a full signing of K_n by assigning signs to all remaining edges, and its ratio persists on the face fixing the new coordinates to zero. Since $M_2 = 1$, this proves the last row. $\square$

In particular $M_7 < M_6$ while $F_7 = F_6 = 3$. An explicit witness is the K_6 signing in Table 3, extended by a new vertex with all incident coefficients positive. At the point whose six old coordinates are $1/2$ and whose new coordinate is zero, its ratio is three; at the full center its ratio is only $21/10$. The finite computation optimizes precisely these listed full-signing problems; it makes no claim about larger n or the optimal constants for arbitrary real coefficients.

3.6 The relation to Schur-multiplier theory

We record the older implication to separate the density order from the explicit elementary constant. For a real matrix M , its Schur multiplier acts by entrywise multiplication, with norm $\|M\|_{\text{Schur}}$ measured in the Euclidean operator norm. Define the real projective norm

$$\pi(M) = \inf \left\{ \sum_k \|u_k\|_\infty \|v_k\|_\infty : M = \sum_k u_k v_k^\top \right\}.$$

The established results of Davidson and Donsig [2, Theorems 1.2 and 2.4] give

$$\|M\|_{\text{Schur}} \leq 2\sqrt{\beta(P)}, \quad \pi(M) \leq K_G \|M\|_{\text{Schur}}, \quad (3.15)$$

for $|M_{ij}| \leq 1$ supported on a pattern P , where K_G is the real Grothendieck constant and

$$\beta(P) = \max_{R, C: |R|+|C|>0} \frac{|P \cap (R \times C)|}{|R| + |C|}.$$

The first bound uses the weighted matrix theorem, which permits continuous row and column bounds. Using the integer pattern theorem instead would introduce rounding. These are established norm inequalities, not statements about sizes of convex extended formulations.

For the symmetric adjacency pattern,

$$|P \cap (R \times C)| \leq |E(G[R \cup C])| + |E(G[R \cap C])| \leq \rho_G(G)(|R| + |C|).$$

An edge is counted at most once unless both endpoints lie in $R \cap C$, when both matrix orientations are counted. Taking $R = C$ equal to a densest set proves $\beta(P) = \rho_G(G)$. For weighted adjacency A , take $M = \text{sign}(A)$, with zeros where A vanishes. The definition of π gives

$$2L_V = \langle A, M \rangle \leq \pi(M) \|A\|_{\infty \rightarrow 1}.$$

For sign vectors u, v , put $p = (u+v)/2$, $q = (u-v)/2$. Symmetry yields $u^\top Av = 2[Q_{a,V}(p) - Q_{a,V}(q)]$. Multiaffinity puts these two values between the vertex extrema on $[-1, 1]^V$, so $\|A\|_{\infty \rightarrow 1} \leq 4R_V$. Consequently

$$L_V \leq 4K_G \sqrt{\rho_G(G)} R_V. \quad (3.16)$$

Apply this on every induced graph to recover the density upper order. The direct random-sign argument supplies the matching bilinear lower order.

Equivalently, let $S(G) = \sup_{a \neq 0} L_V(a) / \|Q_{a,V}\|_\infty$ be the real Sidon constant of the edge-supported degree-two sign characters, with $S(G) = 0$ for edgeless graphs. Because Q has mean zero, $\|Q\|_\infty / 2 \leq R_V \leq \|Q\|_\infty$ and hence $S(G) \leq \Gamma(G) \leq 2S(G)$. For bipartite graphs, flipping all signs in one part negates Q , making its range symmetric and giving $S(G) = \Gamma(G)$. The constant four improves the particular transfer (3.16); it is not asserted to be the best constant available in all prior literature.

4 Universal bounds for positive multilinear functions

Throughout this section the factors are the positive monomials, with equal supports collected. Let R_d be the supremum of T/H over all dimensions, positive multilinear polynomials of maximum degree at most d , finite nonnegative boxes, and points with $H > 0$. Let C_n be the corresponding supremum in n variables, with no additional degree restriction. Fixed coordinates and affine summands may be removed. Proposition 2.4 shows that both suprema can equivalently be taken on the unit cube: expansion does not increase dimension or degree and can only increase the termwise gap. All logarithms without a subscript are natural.

Theorem 4.1 (Sharp leading growth). *As the indicated integer parameter tends to infinity,*

$$R_d \sim \frac{\log d}{\log \log d}, \quad C_n \sim \frac{\log n}{\log \log n}. \quad (4.1)$$

For every $d \geq 2$, put $\Lambda = \max\{16, 1 + \log d\}$ and $c_0 = 1 - e^{-1}$. The explicit finite bound

$$T \leq \frac{24\Lambda}{c_0 \log \Lambda} H \quad (4.2)$$

holds on every finite nonnegative box. On the unit cube the upper bounds are witnessed by distributions defined from the complete marginal vector and degree allowance, independently of the supports and coefficients.

We first exhibit the obstruction to a constant bound, then construct a common distribution that matches its order. The lower family disproves Conjecture 1 in the author manuscript of Luedtke et al. [4, p. 22], which asks for a uniform constant for positive multilinear functions on nonnegative boxes. The signed bilinear examples of Section 3.6 concern a different coefficient class.

4.1 A sparse multiscale obstruction

Theorem 4.2 (Exact dyadic hull gap). *For every integer $L \geq 2$, there is a unit-coefficient polynomial in $n_L = 2^L + L$ variables, with $2^{L+1} - 2$ monomials, maximum degree $d_L = 2^{L-1} + 1$, and an interior point such that*

$$T = L, \quad H = s + (L - s)2^{-s}, \quad (4.3)$$

where $s \in \{1, \dots, L - 1\}$ satisfies

$$(L - s + 1)2^{-(s+1)} \leq 1 \leq (L - s + 2)2^{-s}. \quad (4.4)$$

Either adjacent choice at a cutoff equality gives the same value. The total number of variable occurrences is $L2^L + 2^{L+1} - 2$.

Proof. Put $m = 2^L$. Use leaf variables $z_1, \dots, z_m$ and anchors $a_1, \dots, a_L$. For each j , partition the leaves into 2^j blocks of size $k_j = 2^{L-j}$, and define

$$f_L(a, z) = \sum_{j=1}^L \sum_{B \in \mathcal{P}_j} a_j \prod_{i \in B} z_i, \quad a_j = 2^{-j}, \quad z_i = 1 - 1/m. \quad (4.5)$$

The upper and lower envelopes of each monomial at this point are 2^{-j} and $\max(0, 2^{-j} - k_j/m) = 0$. Thus $T = \text{cav } f_L = L$. The support and occurrence counts follow by summing 2^j and $2^j(k_j + 1)$, respectively.

For an arbitrary binary law with these means, let R count failed leaves and N_j count blocks hit by a failure. Then $\mathbb{E}R = 1$ and $N_j \leq \min(2^j, R)$. The deficiency representation gives

$$H = \max \mathbb{E} \sum_j A_j N_j \leq \sup \mathbb{E} \sum_j A_j \min(2^j, R), \quad \mathbb{E}A_j = 2^{-j}. \quad (4.6)$$

For a fixed law of R , an anchor of mass u maximizes its expectation against any increasing function of R by selecting the upper u -tail, splitting an atom if necessary. Indeed, moving selected mass from a smaller value to a larger unselected value cannot reduce the objective. A common nonincreasing quantile $r(t)$, $0 < t < 1$, therefore makes every anchor $A_j = \mathbf{1}\{t \leq 2^{-j}\}$ optimal simultaneously. Consequently the right side of (4.6) equals the supremum of $\int_0^1 S_{l(t)}(r(t)) dt$, where

$$S_l(r) = \sum_{j=1}^l \min(2^j, r), \quad w_l = \mathbb{P}(l(t) = l) = \begin{cases} 2^{-l-1}, & 0 \leq l < L, \\ 2^{-L}, & l = L. \end{cases}$$

Here r is integer valued, bounded by m , and has integral one.

Define $B_q = (L - q + 2)2^{-q}$. The sequence decreases strictly, with $B_1 \geq 1 \geq B_L$, so an s as in (4.4) exists. For every $r \geq 0$,

$$S_l(r) \leq sr + F_l(s), \quad F_l(s) = \begin{cases} 0, & l \leq s, \\ 2^{l-s+1} - 2, & l > s. \end{cases} \quad (4.7)$$

For $l \leq s$, bound each term by r . For $l > s$, cap the first $l - s$ terms at 2^j and the remaining s terms at r . Integrating yields

$$H \leq s + \sum_l w_l F_l(s) = s + (L - s)2^{-s}.$$

To attain this bound, choose l with probabilities w_l , let $A_j = \mathbf{1}\{j \leq l\}$, and for $q \in \{s, s+1\}$ put

$$R_l^{(q)} = \begin{cases} 0, & l < q, \\ 2^{l-q+1}, & l \geq q. \end{cases}$$

Each anchor has its required mean and $\mathbb{E}R^{(q)} = B_q$. Mix $q = s$ and $q = s+1$ with respective probabilities $(1 - B_{s+1})/(B_s - B_{s+1})$ and $(B_s - 1)/(B_s - B_{s+1})$. This gives $\mathbb{E}R = 1$. For $l > s$ the two failure counts are the endpoints of $[2^{l-s}, 2^{l-s+1}]$, where (4.7) is an equality. For $l = s$ they are 0, 2, also equality points, and for $l < s$ both are zero.

It remains to realize the hit counts, a step requiring nested partitions. Label leaves by L -bit strings, with level- j blocks defined by the first j bits. The first R strings in bit-reversal order have exactly $\min(2^j, R)$ different prefixes of length j : their prefixes are the reversed residues $0, \dots, R-1$ modulo 2^j . Declare these leaves failed and apply a uniform bitwise XOR shift to every label. The shift preserves all hit counts and gives each leaf failure probability R/m , conditional on R . Averaging gives failure mean $1/m$ for every leaf. The constructed law therefore attains the bound in the original polynomial. $\square$

The cutoff conditions imply $s = \log_2 L + O(1)$ and $(L-s)2^{-s} = O(1)$. Thus the ratio in (4.3) is asymptotic to $L/\log_2 L$. The exact formula uses nested dyadic partitions. For arbitrary equal-size partitions the same surrogate proves the weaker but still divergent lower bound

$$\frac{T}{H} \geq \frac{L}{5 + \log_2(1 + (L-1)\log 2)}; \quad (4.8)$$

Appendix B gives its distinct integral proof and the dense symmetric predecessor.

The construction may be made homogeneous of degree d_L : introduce $d_L - 2$ padding variables and multiply a degree- k term by the first $d_L - k$ of them. At padding means one both gaps agree exactly with those above, since all representing laws fix the padding coordinates. Moving them sufficiently close to one makes all means interior with ratio arbitrarily close to (4.3). This adds $O(n_L)$ variables and keeps $O(n_L)$ monomials, but the occurrence count after padding is $O(n_L^2)$, not the original $O(n_L \log n_L)$.

4.2 Harmonic failure distributions

Call a coordinate low when $x_i \leq 1/2$ and high otherwise. Independence handles the terms with zero or at least two low coordinates by Lemma 2.3. For the remaining terms we require failures at several probability scales simultaneously.

Fix a real $M \geq d-1$, and put

$$\Lambda = 1 + \log M, \quad \eta = (d-1)/M \in (0, 1]. \quad (4.9)$$

We use η for this cutoff parameter, to distinguish it from graph density and physical box aspect ratio. Both laws below draw one uniform $U \in (0, 1)$ and set every low coordinate to $\mathbf{1}\{U \leq x_i\}$. In the *threshold law*, a high coordinate with failure mean $p = 1 - x_i$ fails exactly when $U \leq p$. In the *harmonic law*, a coordinate with $p = 0$ never fails; otherwise set

$$h_p = \min(Mp, 1), \quad A_p = 1 + \log(h_p/p), \quad q_p(t) = \frac{\min(1, p/t)}{A_p} \mathbf{1}\{t \leq h_p\}. \quad (4.10)$$

Conditional on $U = t$, high coordinates fail independently with probabilities $q_p(t)$. Since $M \geq 1$, $1 \leq A_p \leq \Lambda$ and $h_p \geq p$. In particular these probabilities are valid, and

$$\int_0^1 q_p(t) dt = \frac{p + p \log(h_p/p)}{A_p} = p.$$

Thus the two laws preserve the full marginal vector, including boundary means, and require no information about a particular term.

Lemma 4.3 (Complete harmonic guarantee). *Consider a term with one low anchor of mean u , and high failure means $p_1, \dots, p_r$, where $r \leq d-1$. Put $t_e = \min(u, \sum_j p_j) > 0$ and $z =$*

$\min(\max_j p_j/t_e, 1)$. The threshold and harmonic deficiencies are at least zt_e and $J_{\Lambda,\eta}(z)t_e$, respectively, where

$$F_{\Lambda,\eta}(v) = 1 - \exp\left(-\frac{1-\eta v}{\Lambda v}\right), \quad J_{\Lambda,\eta}(z) = \int_z^1 F_{\Lambda,\eta}(v) dv, \quad F_{\Lambda,\eta}(0) = 1. \quad (4.11)$$

Proof. Threshold deficiency is $\min(u, \max_j p_j) \geq zt_e$. If $\max_j p_j \geq t_e$, the harmonic assertion is nonnegativity. Otherwise integrate over $\max_j p_j \leq t \leq t_e$. The anchor succeeds there and all $p_j \leq t$. An inactive coordinate has $p_j < t/M$, so the inactive failure mass is at most $rt/M \leq \eta t$. Active mass is consequently at least $\sum_j p_j - \eta t \geq t_e - \eta t$. On active coordinates, $q_{p_j}(t) \geq p_j/(\Lambda t)$. Conditional independence gives union failure probability at least $1 - \exp(-(t_e - \eta t)/(\Lambda t))$. Substitution $v = t/t_e$ proves the claim. Zero term gaps require no division and satisfy every asserted nonnegative bound automatically. $\square$

Theorem 4.4 (Optimized finite harmonic bound). *For the parameters in (4.9), let ζ be the unique solution of $J_{\Lambda,\eta}(\zeta) = \zeta$ in $(0, 1)$. Then*

$$R_d \leq \frac{1}{\zeta} + \frac{2}{c_0}. \quad (4.12)$$

If $w > 0$ solves $we^w = \Lambda e^{-\eta}$ and $w \geq 1$, then

$$\zeta \geq \frac{w - 1/(2w)}{\Lambda}, \quad R_d \leq \frac{\Lambda}{w - 1/(2w)} + \frac{2}{c_0}. \quad (4.13)$$

The parameter M can be optimized over $[d-1, \infty)$ in either bound. For $N = 1 + \log(d-1)$, choosing $M = (d-1)N$ gives, as $d \rightarrow \infty$,

$$R_d \leq \frac{N}{\log N - \log \log N + o(1)} + \frac{2}{c_0}. \quad (4.14)$$

This is an upper certificate, with no matching second-order lower assertion.

Proof. F is continuous and strictly decreasing, positive on $(0, 1)$, and nonnegative at one. Thus J is strictly decreasing and strictly convex, and $J(z) - z$ has opposite signs at zero and one. Its unique root is ζ . Set $a = F(\zeta) \in (0, 1)$. The convex tangent inequality gives

$$J(z) + az \geq J(\zeta) + a\zeta = (1+a)\zeta.$$

Mix the harmonic and threshold laws with weights $1/(1+a)$ and $a/(1+a)$. Lemma 4.3 gives deficiency at least ζt_e for every one-low term. This mixture is optimal for the two *scalar guarantees* $J(z)$ and z : at $z = \zeta$ every convex combination equals ζ . It does not prove optimality over all actual couplings or imply simultaneous attainability of these lower curves by an extremal polynomial. Mix this law and independence with weights proportional to $1/\zeta$ and $2/c_0$. Each term receives at least $t_e/(1/\zeta + 2/c_0)$ from its appropriate component; the other contributions are nonnegative. Proposition 2.2 and then Proposition 2.4 prove (4.12).

For the finite certificate, use $x - x^2/2 \leq 1 - e^{-x} \leq x$ for $x \geq 0$ and, for $0 < z \leq 1$,

$$\int_z^1 (1/v - \eta)^2 dv = 1/z - 1 + 2\eta \log z + \eta^2(1-z) \leq 1/z.$$

The last inequality follows from $2\eta \log z \leq 0$ and $-1 + \eta^2(1-z) \leq 0$. For $0 < A \leq \Lambda$ this gives

$$\Lambda J(A/\Lambda) \geq \log \Lambda - \log A - \eta + \eta A/\Lambda - 1/(2A) \geq \log \Lambda - \log A - \eta - 1/(2A). \quad (4.15)$$

Take $A = w - 1/(2w)$. Since $w \geq 1$, $A \geq 1/2$ and $A < \Lambda$. Using $w + \log w = \log \Lambda - \eta$,

$$A + \log A + \frac{1}{2A} - (w + \log w) = \log \left(1 - \frac{1}{2w^2}\right) + \frac{1}{4Aw^2} \leq 0.$$

The inequality follows from $\log(1 - t) \leq -t$ and $A \geq 1/2$. Hence $J(A/\Lambda) \geq A/\Lambda$; monotonicity of $J(z) - z$ proves (4.13).

For the chosen cutoff, $\Lambda = N + \log N$ and $\eta = 1/N$. The defining identity for w first gives $\log \Lambda - \eta - \log \log \Lambda \leq w \leq \log \Lambda$ for all sufficiently large N . Thus $w/\log N \rightarrow 1$, and substitution back gives $w = \log N - \log \log N + o(1)$. Since $(\log N)^2/N \rightarrow 0$,

$$\frac{N}{\Lambda} \left(w - \frac{1}{2w}\right) = \log N - \log \log N + o(1).$$

This proves (4.14). $\square$

Proof of Theorem 4.1. For its original explicit finite bound, take $\Lambda = \max\{16, 1 + \log d\}$ and $M = e^{\Lambda-1} \geq d$. For a one-low term, put $t_e = \min(u, \sum_j p_j) > 0$. If $\max_j p_j \geq t_e/\sqrt{\Lambda}$, threshold deficiency is at least $t_e/\sqrt{\Lambda} \geq t_e \log \Lambda/\Lambda$; here $\log \Lambda \leq \sqrt{\Lambda}$ follows by maximizing $(\log t)/\sqrt{t}$. Otherwise integrate the harmonic law over $[t_e/\sqrt{\Lambda}, t_e/2]$. Inactive failure mass is at most t , so active mass is at least $t_e/2$. The conditional union probability is at least $c_0 t_e/(2\Lambda t)$, using $1 - e^{-z} \geq c_0 \min(1, z)$. The harmonic deficiency is at least

$$\frac{c_0 t_e}{4\Lambda} (\log \Lambda - 2 \log 2) \geq \frac{c_0 t_e \log \Lambda}{8\Lambda}.$$

An equal mixture of independence, threshold and harmonic laws now gives (4.2), since independence supplies at least $c_0 t_e/2$ on the other two classes. Every distribution is global.

Theorem 4.4 implies $R_d \leq (1 + o(1)) \log d / \log \log d$. For the reverse inequality choose the largest L with $2^{L-1} + 1 \leq d$ in Theorem 4.2. Its degree differs from d by a factor at most two, asymptotically, and its ratio is asymptotic to $L/\log_2 L$. For dimension, $C_n \leq R_n$; choose the largest L with $2^L + L \leq n$ and add unused coordinates. Again the dimensions differ by at most a constant factor. These comparisons prove both equivalences in (4.1) for all integers tending to infinity, not merely for a dyadic subsequence. $\square$

For comparison, the earlier leading-constant mixture has a distinct explicit parameterization. Set $\Lambda = \max\{e^6, 1 + \log d\}$, $M = e^{\Lambda-1}$, $b = \log \Lambda$, $a = \Lambda/b^2$, $\beta = 1/b$, and

$$h = \frac{(1 - 1/b)(1 - 1/(2b^2))(b - 3 \log b)}{\Lambda}. \quad (4.16)$$

Here $h > 0$ because $b \geq 6$ and $b - 3 \log b > 0$. If $\max p_j \geq t_e/a$, threshold deficiency is at least t_e/a . Otherwise, on $[t_e/a, \beta t_e]$, active mass is at least $(1 - \beta)t_e$ and the lower exponent $(1 - \beta)t_e/(\Lambda t)$ is at most $1/b^2$. Integrating $1 - e^{-z} \geq z(1 - 1/(2b^2))$ gives harmonic deficiency at least $h t_e$. Weights proportional to $1/h, a, 2/c_0$ on harmonic, threshold, and independence give $R_d \leq 1/h + a + 2/c_0$, also with leading constant one. The full-curve theorem retains more of the same distributions' information and improves this certificate's lower-order loss.

A useful interpretation is simultaneous interpolation between Fréchet bounds. The global law with $Z = 1/\zeta + 2/c_0$ satisfies, for every $e \subseteq [n]$ with $2 \leq |e| \leq d$,

$$\mathbb{E} \prod_{i \in e} X_i \leq (1 - Z^{-1}) \min_{i \in e} x_i + Z^{-1} \max(0, \sum_{i \in e} x_i - |e| + 1). \quad (4.17)$$

The best fraction guaranteed for every marginal vector is asymptotic to $\log \log d / \log d$: the theorem gives the lower guarantee, and summing any proposed guarantee over the dyadic example gives the reverse order and constant. No convex-envelope evaluation algorithm is inferred.

4.3 Removing coefficients without changing the degree supremum

Theorem 4.5 (Unit coefficients and interior homogeneous examples). *For each fixed $d \geq 2$, the supremum R_d is unchanged by requiring unit coefficients, degree exactly d for every term, and strictly interior unit-cube evaluation points simultaneously. Dimension and support size are unrestricted in this assertion.*

Proof. By positive expansion it suffices to approximate any unit-cube example. Delete affine terms and homogenize with $d - 2$ padding variables as above. At padding means one the gaps are unchanged. Continuity of the finite polyhedral envelopes (Lemma 2.1) permits an interior perturbation preserving any strict target-ratio inequality. Normalize by the largest coefficient. We now have one fixed homogeneous polynomial $f = \sum_{e \in E} a_e \prod_{i \in e} x_i$, $0 < a_e \leq 1$, with n variables, s distinct supports, interior means x , and $H(f; x) > 0$.

Replace coordinate i by m clones $y_{i1}, \dots, y_{im}$, and set

$$\bar{y}_i = m^{-1} \sum_{r=1}^m y_{ir}, \quad F_m(y) = m^d f(\bar{y}), \quad y_{ir} = x_i.$$

There are sm^d distinct candidate clone monomials: each uses one clone from each of its d distinct original groups. At the repeated point,

$$\text{vex } F_m = m^d \text{vex } f, \quad \text{cav } F_m = m^d \text{cav } f, \quad T(F_m) = m^d T(f), \quad H(F_m) = m^d H(f). \quad (4.18)$$

Indeed, any binary clone law induces random group averages $q \in [0, 1]^n$ with $\mathbb{E}q = x$. Conditional independent rounding of the original variables with means q_i has expected polynomial $f(q)$, so every normalized clone expectation is an original feasible expectation. Conversely, setting all clones in a group equal to its original binary coordinate embeds every original law. This proves equality of the entire feasible expectation intervals, and hence of both envelopes. Each clone term has its original list of means, proving the termwise identity.

Independently retain each candidate term arising from e with probability a_e and give it coefficient one; call the result G_m . For N_0 independent random variables with range lengths at most one, the bounded-sum inequality of Hoeffding [3, Theorem 2] gives

$$\mathbb{P}(|\sum_j (Y_j - \mathbb{E}Y_j)| > t) \leq 2e^{-2t^2/N_0}. \quad (4.19)$$

For completeness, if $Y \in [a, b]$, the second derivative of $\log \mathbb{E}e^{\lambda Y}$ is its variance under exponential tilting, at most $(b - a)^2/4$: its squared distance from the interval midpoint is at most that quantity. Integrating twice gives $\mathbb{E}e^{\lambda(Y - \mathbb{E}Y)} \leq e^{\lambda^2(b-a)^2/8}$. Multiplication over independent variables, Markov's inequality, and optimization at $\lambda = 4t/N_0$ prove each tail in (4.19). Thus no concentration result is needed without its hypotheses or proof.

Apply this inequality with $N_0 = sm^d$ at each of the 2^{nm} clone vertices, and once to the termwise gap at the repeated point. The latter is a sum of the same retention indicators with fixed weights in $[0, 1]$. A union bound gives failure probability at most

$$2(2^{nm} + 1) \exp(-2t^2/(sm^d)) \quad (4.20)$$

for the two simultaneous assertions

$$\max_{v \in \{0,1\}^{nm}} |G_m(v) - F_m(v)| \leq t, \quad |T(G_m) - T(F_m)| \leq t.$$

Take $t = Km^{(d+1)/2}$ with $K^2 > sn \log 2/2$. The failure probability tends to zero as $m \rightarrow \infty$, with f, n, s, d fixed. Uniform vertex error bounds the error of each envelope at every marginal vector by t , by comparing expectations under all feasible laws. Thus the hull-gap error is at most $2t$. Since $t/m^d \rightarrow 0$ for $d \geq 2$, successful samples' normalized gaps converge to those of f , and their ratios converge because its hull gap is positive. This proves equality of suprema, not preservation at a fixed dimension or existence of a small sampled support. $\square$

5 Cubic gaps and the role of unequal means

The divergence in Theorem 4.1 requires increasing degree. Degree three already distinguishes positive multilinear functions from the positive bilinear factor-two theorem, however. We prove

$$\frac{1610000}{743033} \leq R_3 \leq \frac{31}{12}. \quad (5.1)$$

The lower endpoint is a bound on a supremum supplied by a sequence of certificates, not a claimed attained finite optimum. We also give exact finite hull values and a homogeneous unit-coefficient example with only 52 variables. The exact value of R_3 remains open.

5.1 Endpoint orientation and a cubic mixture

One particularly simple global law applies in every degree. Draw $U \in [0, 1]$ uniformly and independently orient each coordinate left or right with equal probability. A left coordinate is the indicator of $[0, x_i]$ and a right coordinate of $[1 - x_i, 1]$. Call this law O . Every coordinate has mean x_i .

Proposition 5.1 (Endpoint-orientation bound). *For maximum degree $d \geq 2$, on every nonnegative box,*

$$T \leq \frac{d-1}{1-2^{-(d-1)}} H. \quad (5.2)$$

In particular the constant is $8/3$ for cubics and two for bilinear terms.

Proof. On the unit cube, fix a degree- $(k+1)$ term with minimum mean u at its anchor. Conditional on the anchor orientation, a same-oriented coordinate imposes no further restriction on its success interval. An opposite coordinate excludes an interval of length $a_j = \min(u, 1 - x_j)$ at the opposite end. These excluded intervals are nested. If $a_{(1)} \geq \dots \geq a_{(k)}$, the first selected index has probability 2^{-j} of being j , and hence

$$\mathbb{E}_O D_e = \sum_{j=1}^k 2^{-j} a_{(j)} \geq \frac{1-2^{-k}}{k} \sum_j a_j \geq \frac{1-2^{-k}}{k} \min(u, \sum_j (1-x_j)).$$

The first inequality follows by summing $(2^{-i} - 2^{-j})(a_{(i)} - a_{(j)}) \geq 0$ over pairs, and the second by $\sum_j \min(u, p_j) \geq \min(u, \sum_j p_j)$. The fraction $(1-2^{-k})/k$ decreases with k , since it is the average of the first k entries of $1/2, 1/4, \dots$. Sum the simultaneous guarantees and use Proposition 2.4. $\square$

Theorem 5.2 (Three-law cubic bound). *Every positive multilinear polynomial of degree at most three on a finite nonnegative box satisfies $T \leq (31/12)H$. On the unit cube, the global law*

$$P = \frac{18}{31}O + \frac{6}{31}I + \frac{7}{31}B \quad (5.3)$$

guarantees deficiency at least $12t_e/31$ for every quadratic or cubic term. Here I is independent rounding. For B , draw one uniform U , set each low coordinate to $\mathbf{1}\{U \leq x_i\}$, and, independently conditional on U , let each high coordinate fail with probability $1/2$ when $U \leq 2(1 - x_i)$ and never otherwise. The weights are optimal among fixed mixtures of these three laws for a uniform termwise guarantee.

Proof. The definition of B gives each high coordinate failure mean $1 - x_i$; thus all three laws have the prescribed means. Consider sorted cubic means $u \leq v \leq w$, and set $t = \min(u, 2 - v - w)$. Terms with $t = 0$ need no estimate. Write D_O, D_I, D_B for their expected deficiencies.

If $u \leq v \leq 1/2$, then $t = u$. Opposite anchor and second-coordinate orientations give $D_O \geq u/2$, while $D_I = u(1 - vw) \geq u/2$. These two contributions alone give $12t/31$.

If exactly one coordinate is low, set $a = 1 - v \geq b = 1 - w$. Directly integrating the nested exclusion and failure intervals gives

$$D_O = \frac{1}{2} \min(u, a) + \frac{1}{4} \min(u, b), \quad D_B = \frac{1}{2} \min(u, 2a) + \frac{1}{4} \min(u, 2b).$$

We verify $18D_O + 7D_B \geq 12 \min(u, a + b)$ in all cases. If $a \leq u/2$, the left side is $16a + 8b \geq 12(a + b)$. If $b \geq u/2$, it is at least $18(3u/8) + 7(3u/4) = 12u$. If $b \leq u/2 \leq a \leq u$, it is $9a + 8b + 7u/2$. When $a + b \leq u$, use $3a + 4b \leq 4u - a \leq 7u/2$; when $a + b \geq u$, use $9a + 8b \geq 8u + a \geq 17u/2$. Finally, if $b \leq u/2$ and $a \geq u$, the expression is $25u/2 + 8b \geq 12u$. These cases exhaust $a \geq b \geq 0$, including their boundaries. The I contribution is nonnegative.

If all coordinates are high, put $c = 1 - u \geq a = 1 - v \geq b = 1 - w$. Integrating the B union probabilities $7/8, 3/4, 1/2$ over intervals of lengths $2b, 2(a - b), 2(c - a)$ gives total failure probability $c + a/2 + b/4$. Subtracting the anchor failure probability c yields

$$D_B = D_O = a/2 + b/4 \geq 3(a + b)/8 \geq 3t/8.$$

Independence gives $D_I = u(a + b - ab) \geq 7t/16$. Indeed, set $s = a + b \leq 1$ and use $ab \leq s^2/4$. For $s \leq u$ the normalized expression is at least $u(1 - s/4) \geq u(1 - u/4) \geq 7/16$. For $s \geq u$ it is at least $s - s^2/4 \geq u - u^2/4 \geq 7/16$. Both comparisons use $u > 1/2$ and monotonicity on the unit interval. Thus the mixture fraction is at least $(25/31)(3/8) + (6/31)(7/16) = 12/31$.

For sorted quadratic means $u \leq v$, set $t = \min(u, 1 - v)$; always $D_O = t/2$. If both means are low, $D_I \geq t/2$. If exactly one is low, $D_B = \min(u, 2(1 - v))/2 \geq t/2$. If both are high, $D_B = t/2$ and $D_I = ut \geq t/2$. The resulting mixture fractions are at least $12/31, 25/62$, and $1/2$. This proves the simultaneous guarantee. Summation and positive expansion give the claimed box bound.

To check optimality in the stated family, let the arbitrary weights of (O, I, B) be (w, r, v) with $w + r + v = 1$. The following test means give the listed exact or limiting normalized deficiencies:

means	D_O/t	D_I/t	D_B/t
$(u, 1/2), 0 < u \leq 1/2$	$1/2$	$1/2$	0
$(\varepsilon, 1 - \varepsilon^2, 1 - \varepsilon^2), \varepsilon \downarrow 0$	$3/8$	0	$3/4$
$(u, 1 - u/2, 1 - u/2), u \downarrow 1/2$ from above	$3/8$	$7/16$	$3/8$

In the second row $t = 2\varepsilon^2$ for small ε and the independent fraction is $\varepsilon - \varepsilon^3/2$. In the last row $1/2 < u < 2/3$ ensures $t = u$ and the independent fraction is $u - u^2/4$. Therefore a universal fraction α obeys

$$\alpha \leq \frac{1}{2} - \frac{v}{2}, \quad \alpha \leq \frac{3}{8} + \frac{3v}{8} - \frac{3r}{8}, \quad \alpha \leq \frac{3}{8} + \frac{r}{16}.$$

Average these inequalities with weights $3/31, 4/31, 24/31$. The variable coefficients cancel, giving $\alpha \leq 12/31$. This conclusion concerns fixed-mixture termwise guarantees; it does not identify R_3 , or exclude a stronger coefficient-dependent analysis. $\square$

5.2 A scalar certificate and the strongest limiting lower bound

For a set Q of variables, write $E_k(Q) = \sum_{S \subseteq Q, |S|=k} \prod_{i \in S} x_i$. At a binary vertex this is $\binom{K}{k}$ when K counts successes in Q . Count-polynomial identities below are vertex identities, not substitutions defining the multilinear functions at continuous points.

Lemma 5.3 (Cubic scalar minorant with positive slack). *On $[0, 1]^3$, let*

$$\begin{aligned} F(a, b, c) &= 6c^3 + 27bc^2 + 18b^2 + 30ac + 20ab + 9a^2, \\ \ell(a, b, c) &= 37a + \frac{79}{2}b + 38c - \frac{103}{3}. \end{aligned}$$

Then $F \geq \ell + \delta_$, where $\delta_* = 901/120000$.*

Proof. For fixed c , the Hessian of $h = F - \ell$ in (a, b) is $\begin{pmatrix} 18 & 20 \\ 20 & 36 \end{pmatrix}$, positive definite with determinant 248. For $0 \leq c \leq 3/10$, set $a = 1$ and $b_*(c) = 13/24 - 3c^2/4 \in (0, 1)$. There $\partial_b h = 0$ and $\partial_a h = -49/6 + 30c - 15c^2 \leq -31/60 < 0$. The convex first-order inequality proves that this is the constrained minimum on the (a, b) square: $(a - 1)\partial_a h \geq 0$ for $a \leq 1$. Its value is

$$p(c) = \frac{-972c^4 + 576c^3 + 1404c^2 - 768c + 101}{96}.$$

For $3/10 \leq c \leq 1$, minimizing over unrestricted real a, b is a valid lower bound. Completing the square gives

$$r(c) = \frac{-78732c^4 + 212256c^3 - 203796c^2 + 82032c - 11275}{2976}.$$

For example, with $v = 30c - 37$, $w = 27c^2 - 79/2$ and $q = 6c^3 - 38c + 103/3$, this value is $q - (18v^2 - 20vw + 9w^2)/248$, which verifies the elimination directly.

Table 1 supplies exact degree-four Bernstein representations. A row with interval $[l, u]$, denominator D , and numerators $v_0, \dots, v_4$ means

$$p(c) \text{ or } r(c) = D^{-1} \sum_{i=0}^4 v_i \binom{4}{i} t^i (1-t)^{4-i}, \quad t = (c-l)/(u-l).$$

These are rational polynomial identities, verified by expanding in t . The five intervals cover the required ranges. The Bernstein basis is nonnegative and sums to one, and the minimum of all 25 coefficients v_i/D is $1802/240000 = \delta_*$. Thus both lower polynomials are at least δ_* wherever used, proving the claim. $\square$

Polynomial	Interval	D	$(v_0, v_1, v_2, v_3, v_4)$
p	$[0, 1/5]$	60000	(63125, 39125, 20975, 9395, 4133)
p	$[1/5, 3/10]$	240000	(16532, 6008, 1802, 3788, 11597)
r	$[3/10, 1/2]$	7440000	(215357, 1285415, 1434125, 1250375, 1008125)
r	$[1/2, 3/4]$	190464	(25808, 18056, 7964, 9230, 15869)
r	$[3/4, 1]$	190464	(15869, 22508, 34520, 45920, 31040)

Table 1: Exact scalar positivity certificate for Lemma 5.3.

Theorem 5.4 (Analytic cubic lower family). *For each multiple $m \geq 9$ of nine, take three groups U, V, W of m variables and write A, B, C for their sums. Define*

$$f_m = 2E_3(W) + 3BE_2(W) + 2mE_2(V) + \frac{5m}{3}AC + \frac{10m}{9}AB + mE_2(U). \quad (5.4)$$

At group means $1/4, 1/2, 3/4$, respectively,

$$\frac{T(f_m)}{H(f_m)} \geq \frac{161/4 - 135/(4m) + 6/m^2}{223/12 - \delta_* + 135/(4m) + 9/m^2}. \quad (5.5)$$

Consequently $R_3 \geq 1610000/743033$.

Proof. All coefficients in (5.4) are positive integers. At binary counts put $a = A/m, b = B/m, c = C/m$. Expansion of binomial coefficients gives the exact identity

$$\frac{18f_m}{m^3} = F(a, b, c) - \frac{18c^2 + 27bc + 18b + 9a}{m} + \frac{12c}{m^2} \geq F(a, b, c) - 72/m. \quad (5.6)$$

Every feasible binary law has expected normalized counts $(1/4, 1/2, 3/4)$. Lemma 5.3 therefore gives

$$\frac{18}{m^3} \text{vex } f_m \geq 139/6 + \delta_* - 72/m.$$

Counting supports and using (2.6) yields

$$\frac{18}{m^3} \text{cav } f_m = 167/4 - 153/(4m) + 9/m^2, \quad (5.7)$$

$$\frac{18}{m^3} T(f_m) = 161/4 - 135/(4m) + 6/m^2. \quad (5.8)$$

For clarity, the six scaled upper contributions in the displayed family order are

$$\frac{9}{2} - \frac{27}{2m} + \frac{9}{m^2}, \quad \frac{27}{2} - \frac{27}{2m}, \quad 9 - \frac{9}{m}, \quad \frac{15}{2}, \quad 5, \quad \frac{9}{4} - \frac{9}{4m}.$$

Only $2E_3(W)$ has a positive termwise lower envelope; its scaled value is $3/2 - 9/(2m) + 3/m^2$. This proves the two exact formulas. Subtraction bounds the scaled hull gap by the denominator in (5.5). It is positive for $m \geq 9$, and the actual hull gap is positive as well: independence gives strictly smaller expectations than common-threshold rounding for every positive nonlinear term at interior means. Division is therefore valid. Taking arbitrarily large multiples of nine yields

$$R_3 \geq \frac{161/4}{223/12 - 901/120000} = \frac{4830000}{2229099} = \frac{1610000}{743033}.$$

Dropping the positive slack recovers the weaker $483/223$; the same certificate proves the stronger reduced fraction above. Neither minorant is asserted to be attained in expectation, and neither limiting bound asserts an attained finite gap ratio or convergence of the actual family ratios to that lower endpoint. At $m = 36$, even dropping δ_* from (5.5) gives $T/H \geq 16985/8436 > 2$; retaining the positive slack makes this finite bound strictly stronger. $\square$

5.3 Exact finite cubic witnesses

Finite examples permit a stronger type of statement: matching primal and dual certificates determine both envelopes exactly. The three-group polynomials in Table 2 have group sizes m and means $(1/4, 1/2, 3/4)$, and take the form

$$f = c_1 E_3(W) + c_2 B E_2(W) + c_3 E_2(V) + c_4 AC + c_5 AB + c_6 E_2(U). \quad (5.9)$$

Proposition 5.5 (Exact finite values). *The envelope and ratio values in Table 2 are exact. The 18- and 24-variable polynomials have 212 and 464 monomials, respectively. Complete rational primal and dual certificates, and a short executable integer/rational checker, appear in Appendix C.*

Proof. At binary counts $A, B, C \in \{0, \dots, m\}$ the polynomial is

$$\Phi_m(A, B, C) = c_1 \binom{C}{3} + c_2 B \binom{C}{2} + c_3 \binom{B}{2} + c_4 AC + c_5 AB + c_6 \binom{A}{2}.$$

The appendix certifies an affine minorant at every count state and a probability distribution of count states with means $(m/4, m/2, 3m/4)$ attaining it. Conditional uniform subsets in each group realize every individual marginal, so these give matching lower-envelope certificates. The upper and termwise lower values follow directly from the six orbit sizes and monomial bounds in the appendix. Subtracting yields the ratios. $\square$

n	$(c_1, \dots, c_6)$	$\text{cav } f$	$\sum_e \text{vex } f_e$	$\text{vex } f$	T/H
18	(2, 3, 9, 10, 7, 7)	1647/4	10	2750/13	20891/10411
24	(2, 3, 13, 12, 8, 7)	971	28	3572/7	6601/3225
192	(2, 3, 120, 105, 70, 63)	587944	20832	34172072/105	7443345/3445256

Table 2: Exact cubic witnesses, all with ratio above two.

The 24-variable example also gives a quantitative homogeneous interior variant. Write it as $f = P + Q$, where P is cubic and Q quadratic, and put $g(x, z) = P(x) + zQ(x)$ with $z = 999/1000$. All 25 coordinates are interior and all 464 monomials are cubic, with integer coefficients at most 13. Their upper and lower termwise values are unchanged, so $\text{cav } g = 971$ and $T(g) = 943$. The sum of quadratic coefficients is

$$13 \binom{8}{2} + 12 \cdot 8^2 + 8 \cdot 8^2 + 7 \binom{8}{2} = 1840.$$

Thus for any feasible law, $\mathbb{E}g \geq \text{vex } f - 1840/1000$, regardless of dependence involving Z . Consequently

$$H(g) \leq 3225/7 + 46/25 = 80947/175, \quad \frac{T(g)}{H(g)} \geq \frac{165025}{80947} > 2. \quad (5.10)$$

This is a certified bound, rather than an exact new hull value.

For completeness, it yields a finite nonconstructive unit-coefficient example through Theorem 4.5. Normalize by 13, clone each variable $m = 1000$ times, and take error allowance $t = m^3/10$. The failure probability in (4.20) is at most

$$2(2^{25m} + 1) \exp(-m^3/23200) < 1. \quad (5.11)$$

Indeed its logarithm is at most $(25m + 2) \log 2 - m^3/23200 < 0$ at $m = 1000$, even with $\log 2 < 1$. Since $943 - 2(80947/175) = 3131/175$, every successful sample satisfies

$$T(G_m) - 2H(G_m) \geq m^3 \left(\frac{3131}{2275} - \frac{1}{2} \right) > 0. \quad (5.12)$$

It is nonempty and has positive hull gap at its interior point. This proves existence in 25,000 variables with $464m^3$ candidate monomials; no successful sample is claimed to be listed. The explicit construction below is much smaller, while coefficient removal remains useful for approximating an arbitrary fixed-degree ratio.

5.4 Two means and an explicit homogeneous unit-coefficient example

Theorem 5.6 (Two-level cubic family). *For each positive multiple m of four, take two groups U, W of m variables with means $1/2, 3/4$, and define*

$$f_m = AE_2(W) + \frac{5m}{4} E_2(U), \quad A = \sum_{i \in U} x_i. \quad (5.13)$$

Then

$$\frac{243}{115}(1 - 1/m) \leq \frac{T(f_m)}{H(f_m)} \leq \frac{243}{115}. \quad (5.14)$$

For $m = 4, 8, 12, 16$ its exact ratios are, respectively, $27/16, 21/11, 99/50, 135/67$. Among positive multiples of four, this family's ratio exceeds two if and only if $m \geq 16$. There is an explicit polynomial with 52 variables and 4,320 distinct homogeneous cubic monomials, every coefficient one, at an interior point with ratio at least $2700/1343 > 2$.

Proof. For real $a, c \in [0, 1]$, set $F_2(a, c) = ac^2 + 5a^2/4$ and $\ell_2(a, c) = 11a/6 + 20c/27 - 95/108$. The identity

$$F_2 - \ell_2 = \frac{5}{4} \left(a - \frac{11 - 6c^2}{15} \right)^2 + \frac{(1 - c)(3c - 2)^2(3c + 7)}{135} \quad (5.15)$$

proves $F_2 \geq \ell_2$ on the square. Equality holds at $(1/3, 1)$ and $(5/9, 2/3)$; their mixture with weights $1/4, 3/4$ has mean $(1/2, 3/4)$ and expected value $16/27$. Thus the scalar envelope value at that mean is exactly $16/27$.

At binary counts $A = ma, C = mc$,

$$\frac{2f_m}{m^3} = F_2(a, c) - \frac{ac + (5/4)a}{m}.$$

Every feasible law has $\mathbb{E}a = 1/2$ and $\mathbb{E}c = 3/4$, and $ac \leq a$. It follows that $2 \text{vex } f_m/m^3 \geq 16/27 - 9/(8m)$. Each monomial has upper envelope $1/2$ and lower envelope zero; therefore

$$\frac{2}{m^3} \text{cav } f_m = \frac{2}{m^3} T(f_m) = \frac{9}{8}(1 - 1/m). \quad (5.16)$$

Subtraction gives $2H(f_m)/m^3 \leq 115/216$. For the reverse bound, choose one of the two equality atoms above, then sample all coordinates independently conditional on its two probabilities. Distinct variables in each term have conditional product expectation, so $2\mathbb{E}[f_m \mid a, c]/m^3 = (1 - 1/m)F_2(a, c)$. This is a feasible law and gives $2 \text{vex } f_m/m^3 \leq (1 - 1/m)16/27$. Hence

$$\frac{115}{216}(1 - 1/m) \leq \frac{2H(f_m)}{m^3} \leq \frac{115}{216}.$$

Together with (5.16) this proves (5.14), including positive denominators and convergence of the actual ratios to 243/115. For example, $m = 20$ gives the lower bound $4617/2300 > 2$ in 40 variables.

At $m = 16$, the exact integer-grid certificate is

$$A \binom{C}{2} + 20 \binom{A}{2} \geq 214A + \frac{177}{2}C - 1686 \quad (A, C \in \{0, \dots, 16\}). \quad (5.17)$$

Appendix C supplies its integer verification. Uniform success subsets of sizes 8, 12 give equality, with value 1088. The concave envelope and termwise gap are 2160, so $H = 1072$ and $T/H = 135/67$.

For $m = 4, 8, 12$, Table 6 in Appendix C gives matching affine minorants and uniform fixed-count laws, proving the three exact ratios stated above. All three are below two. For every $m \geq 20$, the lower bound in (5.14) is at least $4617/2300 > 2$, since $1 - 1/m$ is increasing. Together with $m = 16$, these facts prove the if-and-only-if assertion among positive multiples of four in (5.13); they make no minimum-dimension claim over other families.

Introduce 20 distinct coordinates z_k and define explicitly

$$g = AE_2(W) + \left(\sum_{k=1}^{20} z_k \right) E_2(U), \quad x_U = 1/2, \quad x_W = 3/4, \quad z_k = 999/1000. \quad (5.18)$$

There are $16 \binom{16}{2} + 20 \binom{16}{2} = 4320$ distinct cubic supports, all with coefficient one. Every term still has upper envelope $1/2$ and lower envelope zero, so $\text{cav } g = T(g) = 2160$. At binary vertices,

$$g = f_{16} - \left(\sum_k (1 - z_k) \right) E_2(U), \quad 0 \leq E_2(U) \leq 120.$$

Every feasible law projects to the original means, giving $\text{vex } g \geq 1088 - 120(20/1000) = 1088 - 12/5$. Thus $0 < H(g) \leq 5372/5$ and $T(g)/H(g) \geq 2700/1343$. Positivity of H follows by comparing independent and common-threshold laws at these interior means. $\square$

The two-level examples show that means in the upper half of the unit cube do not force factor two. Small upward perturbations of the $1/2$ means preserve a strict ratio above two by continuity, so all means can be strictly above $1/2$. No quantitative perturbation radius or minimal dimension claim is implied. The next result explains why at least two distinct means are needed on the unit cube.

6 Equal means and classical symmetric envelopes

The envelope of the complete elementary symmetric polynomial is classical. Sherali's equation (13) and Theorem 3 [6, pp. 252–253] give, on the entire unit cube,

$$\text{vex } E_d(x) = \max \left(0, \max_{k=d-1, \dots, n-1} \left\{ \binom{k}{d-1} \sum_i x_i - (d-1) \binom{k+1}{d} \right\} \right). \quad (6.1)$$

The following extremal gap formulation is a consequence of that symmetric-envelope machinery. We give a direct proof at equal means to make the common-law guarantee explicit.

Theorem 6.1 (Exact equal-mean formula). *Fix $n \geq 2$ and $0 < u < 1$. Put $b = \lfloor nu \rfloor$, $\theta = nu - b$, and, for $2 \leq d \leq n$,*

$$t_d(u) = \min\{u, (d-1)(1-u)\}, \quad q_{n,d}(u) = \frac{(1-\theta) \binom{b}{d} + \theta \binom{b+1}{d}}{\binom{n}{d}},$$

with $\binom{k}{d} = 0$ for integers $0 \leq k < d$. The maximum of T/H over positive multilinear polynomials with a nonzero nonlinear part at $x = (u, \dots, u)$ is

$$R_n^{\text{eq}}(u) = \max_{2 \leq d \leq n} \frac{t_d(u)}{u - q_{n,d}(u)}. \quad (6.2)$$

A maximum-degree restriction $2 \leq D \leq n$ restricts this maximum to $d \leq D$. Each candidate ratio is attained by E_d .

Proof. Draw $K = b$ or $b + 1$ with probabilities $1 - \theta, \theta$, then choose a uniformly random K -element success subset. Each mean is u and each degree- d product expectation is $q_{n,d}(u)$. This single law gives

$$H(f; u\mathbf{1}) \geq \sum_e a_e(u - q_{n,|e|}(u)), \quad T(f; u\mathbf{1}) = \sum_e a_e t_{|e|}(u).$$

The quotient is bounded by the largest term ratio, provided its denominators are positive. To see positivity and attainment, the binary value of E_d is $\binom{K}{d}$, whose first forward difference is $\binom{K}{d-1}$ and is nondecreasing. Its linear interpolation on integer counts is convex. Jensen's inequality shows that any integer-valued K with mean nu has expected value at least the adjacent-count interpolation. Our uniform-subset law attains this value with all individual means correct, establishing $\text{vex } E_d = \binom{n}{d} q_{n,d}$. Its upper envelope is $u \binom{n}{d}$. Comparing the minimum with the independent law, whose product expectation is u^d , gives $q_{n,d} \leq u^d < u$. Thus the denominators are positive and every claimed candidate is attained. The maximum is attained by a maximizing degree. $\square$

Corollary 6.2 (Dimension-free equal-mean supremum). *For $0 < u < 1$ the exact supremum over dimensions, and the limit of $R_n^{\text{eq}}(u)$ as $n \rightarrow \infty$, is*

$$R_\infty^{\text{eq}}(u) = \max_{k \geq 1} \frac{\min\{1, k(1-u)/u\}}{1 - u^k} \leq 2. \quad (6.3)$$

Writing $r_u = u/(1-u)$, a maximizing k is one of $\max(1, \lfloor r_u \rfloor)$ and $\max(1, \lceil r_u \rceil)$. For $u \leq 1/2$ the value is $1/(1-u)$. The uniform constant two is sharp.

Proof. The comparison $q_{n,d} \leq u^d$ gives the upper bound by the displayed maximum after writing $k = d - 1$. For fixed d , $q_{n,d}(u) \rightarrow u^d$ as $n \rightarrow \infty$, since both adjacent counts divided by n tend to u . Thus any fixed-degree candidate is approached by E_d in increasing dimension.

For $k \geq r_u$ the expression is $1/(1-u^k)$, decreasing with k . For $k \leq r_u$ it equals $k/(u+u^2+\dots+u^k)$, increasing with k because the average of that decreasing sequence decreases. Hence a finite maximizer occurs at one of the stated two integers. Its fixed-degree limit proves both the supremum and convergence assertions.

Set $t = k(1-u)/u$. Bernoulli's inequality gives $u^{-k} = (1 + (1-u)/u)^k \geq 1 + t$, and therefore

$$\frac{\min(1, t)}{1 - u^k} \leq \min(1, t) \frac{1+t}{t} \leq 2.$$

When $u \leq 1/2$ the maximizing integer is $k = 1$, giving $1/(1-u)$. At $u = 1/2$, the complete bilinear polynomial for even n has ratio $2(n-1)/n$, tending to two. This is the classical positive bilinear sharpness example, now viewed within the equal-mean formula. $\square$

These exact formulas concern equal normalized means on the unit cube. Positive expansion transfers the degree upper bounds to nonnegative boxes, but does not identify an original positive-box factorization with the elementary symmetric unit-cube model. In particular, the precise finite equal-mean optimization above makes no such additional box claim.

A Exhaustive verification of the finite signing constants

This appendix gives the finite computational proof used in Proposition 3.6. Label vertices $0, \dots, n-1$. For vertex signs t_i , switching replaces a_{ij} by $a'_{ij} = a_{ij}t_it_j$. The identity $Q_{a'}(s) = Q_a(t \cdot s)$ shows that switching preserves both extrema of Q , and hence R_V . Taking $t_0 = 1$ and $t_j = a_{0j}$ makes every coefficient incident to zero positive. This normalized representative is unique within each switching class: a switching that keeps those coefficients positive has $t_j = t_0$ for all j , so changes no edge. Thus it suffices to enumerate the $2^{\binom{n-1}{2}}$ choices of negative edges among vertices $1, \dots, n-1$. There are respectively 2, 8, 64, 1024, 32768 representatives for $n = 3, \dots, 7$.

Encode a cut by the subset on side one, always leaving vertex zero on side zero. Complementation changes no cut, so the 2^{n-1} such subsets include every cut. If a cut has c crossing edges, of which h are negative, its weight is exactly $c - 2h$. The following complete Python 3.10+ program computes the first row of Proposition 3.6 using unbounded integers. The operation `bit_count()` counts set bits in an integer; each free-edge mask records which edges cross the cut. No floating-point arithmetic or solver is used.

```
from itertools import combinations
def min_range(n):
    edges = list(combinations(range(n), 2))
    free = [(i, j) for i, j in edges if i != 0]
    cuts = []
    for side in range(1 << (n - 1)):
        side <= 1 # Vertex zero stays on side zero.
        cross = lambda i, j: ((side >> i) ^ (side >> j)) & 1
        size = sum(cross(i, j) for i, j in edges)
        mask = sum(cross(i, j) << k
                    for k, (i, j) in enumerate(free))
        cuts.append((size, mask))
    best = len(edges) + 1
    for negative in range(1 << len(free)):
        weights = [size - 2 * (negative & mask).bit_count()
                    for size, mask in cuts]
        best = min(best, max(weights) - min(weights))
    return best
result = [min_range(n) for n in range(2, 8)]
assert result == [1, 2, 4, 4, 5, 8]
print(result)
```

Every signed cut weight lies between $-\binom{n}{2}$ and $\binom{n}{2}$; its range is at most $\binom{n}{2}$, because two cut weights differ by a sum of at most that many coefficients in $\{-1, 1\}$. The initial value `len(edges) + 1` therefore exceeds every candidate range. The two loops cover all normalized signings and all cuts, so their minimum is exactly $\min_a R_V(a)$, rather than merely an upper bound supplied by the best witness found. This is an exhaustive finite proof, not an analytic formula for general n .

The extrema in Table 3 can be checked by direct evaluation of Q on the 2^{n-1} vertex-sign vectors with $s_0 = 1$. They give $R_V = (\max Q - \min Q)/2$ equal to the computed minimum in each case. Extending the listed K_6 witness by vertex six and positive new edges gives $\min Q = -9$, $\max Q = 11$ at the full center, hence ratio 21/10; its original six-vertex face retains ratio three.

The accompanying script `verification/check_complete_signings.py` implements the same integer enumeration, records a histogram of all candidate ranges, and checks the displayed witnesses by direct evaluation of Q on every induced face. Running it with Python 3.10+ writes `verification/`

n	Negative edges; all other edges have coefficient +1	$\min Q$	$\max Q$
3	$\emptyset$	-1	3
4	$\emptyset$	-2	6
5	12, 14, 23	-4	4
6	14, 15, 23, 25, 34	-5	5
7	12, 13, 16, 23, 25, 34	-7	9

Table 3: Attaining full-center signings, with vertices numbered from zero. An entry ij denotes the unordered pair $\{i, j\}$.

`complete_signings.json`, including the witnesses, counts, exact rational ratios, and the extended six-vertex example.

B Distinct multiscale arguments and finite refinements

B.1 Arbitrary partitions and the dense predecessor

The exact dyadic formula in Theorem 4.2 relies on a simultaneous prefix-hitting construction. A separate logarithmic estimate works for arbitrary partitions in (4.5).

Proposition B.1 (Partition-independent estimate). *For the polynomial and means in (4.5), with any choice of equal-size level partitions,*

$$0 < H \leq 5 + \log_2(1 + (L - 1) \log 2), \quad T = L.$$

Proof. Use the quantile surrogate from (4.6), writing $r(t)$ for the nonincreasing failure-count quantile and $\delta = 2^{-L}$. The interval $0 < t \leq \delta$ contributes at most $\delta \sum_{j=1}^L 2^j < 2$, and $t > 1/2$ contributes zero. For $\delta < t \leq 1/2$, let l be the largest integer with $t \leq 2^{-l}$. Write $\log_2^+ v = \max\{0, \log_2 v\}$ for $v > 0$. Then $2^l \leq 1/t$, and for $r > 0$,

$$\sum_{j=1}^l \min(2^j, r) \leq r \left(3 + \log_2^+ \frac{1}{tr} \right). \quad (\text{B.1})$$

If $r \geq 2^l$, the sum is less than $2r$. If $r < 2$, it is lr and $l \leq 1 + \log_2(2^l/r)$. Otherwise, with $h = \lfloor \log_2 r \rfloor$, the first h capped terms sum to less than $2r$ and the others sum to $(l - h)r$; use $h > \log_2 r - 1$. This proves (B.1), interpreted as zero at $r = 0$.

Put $D = (\delta, 1/2)$ and $M_0 = \int_D r(t) dt \leq 1$. As $\log_2^+ v \leq \log(1 + v)/\log 2$, the bound becomes

$$H \leq 2 + 3M_0 + \frac{1}{\log 2} \int_D r(t) \log \left(1 + \frac{1}{tr(t)} \right) dt.$$

For $M_0 > 0$, Jensen's inequality under the probability measure $r(t) dt/M_0$ gives

$$\int_D r(t) \log \left(1 + \frac{1}{tr(t)} \right) dt \leq M_0 \log \left(1 + \frac{(L - 1) \log 2}{M_0} \right) \leq \log(1 + (L - 1) \log 2).$$

The final step uses that $M \log(1 + B/M)$ increases for $M > 0$, since $\log(1 + s) \geq s/(1 + s)$. At $M_0 = 0$ the integral is zero. This proves the upper bound. Independent rounding and common-threshold rounding have distinct expected polynomial values at these interior means, so $H > 0$. $\square$

A dense symmetric construction gives a different exact reduction. With $m = 2^L$, $u_j = 2^{-j}$, $k_j = mu_j$, define

$$\tilde{f}_L(a, z) = \sum_{j=1}^L \frac{1}{u_j \binom{m}{k_j}} \sum_{|K|=k_j} a_j \prod_{i \in K} z_i, \quad a_j = u_j, \quad z_i = 1 - 1/m. \quad (\text{B.2})$$

Again $T = \text{cav } \tilde{f}_L = L$. Conditional on r failed leaves, the fraction of size- k_j subsets hit is

$$g_j(r) = 1 - \frac{\binom{m-r}{k_j}}{\binom{m}{k_j}} \leq \min(1, u_j r).$$

The inequality is the union bound: a uniform size- k_j subset contains each specified failed leaf with probability $k_j/m = u_j$. Thus the deficiency is $\sum_j A_j g_j(R)/u_j$, bounded by the same surrogate $\sum_j A_j \min(2^j, R)$ and consequently by Proposition B.1.

In this dense model the hull gap itself is exactly the optimum of the finite rational program

$$\begin{aligned} \max \quad & \sum_{j=1}^L \sum_{r=0}^m z_{jr} g_j(r) / u_j \\ \text{subject to} \quad & p_r \geq 0, \quad 0 \leq z_{jr} \leq p_r, \\ & \sum_{r=0}^m p_r = 1, \quad \sum_{r=0}^m r p_r = 1, \quad \sum_{r=0}^m z_{jr} = u_j \quad (j = 1, \dots, L). \end{aligned} \quad (\text{B.3})$$

Every feasible binary law supplies $p_r = \mathbb{P}(R = r)$ and $z_{jr} = \mathbb{P}(A_j = 1, R = r)$. Conversely draw R with probabilities p_r , choose a uniform r -subset of failed leaves, and, conditional on $R = r$, choose anchors independently with probabilities z_{jr}/p_r when $p_r > 0$. These choices realize all means and the objective; zero-mass states can be ignored. Thus (B.3) is an exact reduction, not a formula for the sparse model. No floating-point values from this program are needed for the divergent bound.

B.2 Dyadic conditional-failure rounding and its tail lemma

The following proof uses finitely many dyadic scales rather than a harmonic kernel. Though its $O(\log d)$ bound is weaker, its tail argument is distinct.

Lemma B.2 (Tail integral). *For a finite collection $p_j \geq 0$, let $N(s) = |\{j : p_j \geq s\}|$ and $S = \sum_j p_j$. For $u > 0$,*

$$\int_0^u \min\{N(s), u/s\} ds \geq \min(u, S). \quad (\text{B.4})$$

Proof. If $S \leq u$, then every $p_j \leq u$ and $\int_0^u N(s) ds = S$. Monotonicity gives $sN(s) \leq \int_0^s N(t) dt \leq S \leq u$, so the integrand equals $N(s)$ throughout. If $S > u$, then $\int_0^u N(s) ds \geq u$: either a $p_j \geq u$, or that integral equals S . By continuity choose $v \leq u$ with $\int_0^v N(s) ds = u$. For $s \leq v$ the same monotonicity bound gives $sN(s) \leq u$. Integrating on $(0, v)$ therefore gives u . $\square$

Proposition B.3 (Dyadic degree bound). *Let $K = 1 + \lfloor \log_2(d-1) \rfloor$ for $d \geq 2$. Every positive multilinear polynomial of maximum degree at most d on a nonnegative box satisfies $T \leq 2(K+1)H/c_0$.*

Proof. For each scale $B \in \{1, 2, \dots, 2^{K-1}\}$ restricted to powers of two, draw a common uniform U . Low coordinates succeed when $U \leq x_i$. A high coordinate with failure mean $p > 0$ has $h = \min(Bp, 1)$ and, conditional on U , fails independently with probability p/h when $U \leq h$, and

never otherwise. Coordinates with $p = 0$ never fail. The marginal failure is $h(p/h) = p$ and the probabilities are valid because $B \geq 1$.

For a one-low term with anchor mean $u > 0$, let $p_1, \dots, p_r$ be its high failure means. At $U = t \leq u$, each $p_j \geq t/B$ is active and has conditional failure probability at least $1/B$. Conditional independence and $1 - e^{-z} \geq c_0 \min(1, z)$ give deficiency

$$G_B \geq c_0 \int_0^u \min(1, N(t/B)/B) dt = c_0 \int_0^{u/B} \min(B, N(s)) ds.$$

For $0 < s \leq u$ with $N(s) > 0$, choose the largest power of two $B \leq \min(N(s), u/s)$. It is in the allowed set, since $N(s) \leq r \leq d-1$, and exceeds half this minimum. Therefore

$$\sum_B \mathbf{1}\{s \leq u/B\} \min(B, N(s)) \geq \frac{1}{2} \min(N(s), u/s).$$

Integrating and invoking Lemma B.2 yields $\sum_B G_B \geq c_0 \min(u, S)/2 = c_0 t_e/2$. Terms with $u = 0$ have zero gap. Independence supplies the same guarantee for terms with zero or at least two low coordinates by Lemma 2.3. The uniform mixture of the K scale laws and independence thus guarantees $c_0 t_e/(2(K+1))$ for every term. Sum and apply Proposition 2.4. $\square$

B.3 The simplest harmonic cutoff and its reciprocal correction

The general finite theorem permits $M = d-1$, so $\eta = 1$ and $\Lambda = 1 + \log(d-1)$. Let z_Λ solve $J_{\Lambda,1}(z_\Lambda) = z_\Lambda$ and put $\alpha = \Lambda z_\Lambda$, $\ell = \log \Lambda$. For $\ell \geq 4$,

$$z_\Lambda \geq \frac{\ell - \log \ell - 2}{\Lambda}. \quad (\text{B.5})$$

Indeed $A = \ell - \log \ell - 2 \geq 1/2$ and $A \leq \ell < \Lambda$. Equation (4.15) with $\eta = 1$ gives $\Lambda J(A/\Lambda) \geq \ell - \log \ell - 1 - 1/(2A) \geq A$. Monotonicity of $J(z) - z$ proves the finite bound.

The integrated linear upper bound and quadratic lower bound imply

$$\ell - \log \alpha - 1 - \frac{1}{2\alpha} \leq \alpha \leq \ell - \log \alpha - 1 + \frac{\alpha}{\Lambda}.$$

Together with (B.5), these show $\alpha/\ell \rightarrow 1$: once $\alpha \geq 1$, the upper bound gives $\alpha \leq \ell/(1 - 1/\Lambda)$, while the finite lower bound is $\ell - \log \ell - 2$. Substitution yields

$$\Lambda z_\Lambda = \log \Lambda - \log \log \Lambda - 1 + o(1). \quad (\text{B.6})$$

The constant -1 belongs to this fixed cutoff, whereas the tuned cutoff in (4.14) removes that constant after expressing the bound with numerator N .

There is also a reciprocal refinement. If $w e^w = \Lambda/e$, then

$$\alpha = w - \frac{1}{2w} + O(w^{-2}) \quad (\Lambda \rightarrow \infty). \quad (\text{B.7})$$

To prove it, integrate the cubic Taylor bound for $1 - e^{-x}$ with $x = (1-v)/(\Lambda v)$. After multiplication by Λ , the error beyond the quadratic term is at most

$$\Lambda \int_{\alpha/\Lambda}^1 \frac{x^3}{6} dv \leq \frac{1}{12\alpha^2}.$$

The exact quadratic integral is $\int_z^1 (1/v - 1)^2 dv = 1/z + 2 \log z - z$. Its omitted terms, and the linear endpoint term, are $O(\log \Lambda/\Lambda) = o(\alpha^{-2})$ because $\alpha \sim \ell$. Consequently

$$\alpha + \log \alpha = \log \Lambda - 1 - \frac{1}{2\alpha} + O(\alpha^{-2}).$$

Compare this with $w + \log w = \log \Lambda - 1$. Since $\alpha \sim w$, the mean value theorem for $t + \log t$ first gives $\alpha - w = O(1/w)$ and, after substitution, gives (B.7). This refines the scalar certificate only; it is not a second-order lower bound on R_d .

B.4 A second rational two-level parameter choice

A related finite variant illustrates that the two-level mechanism is not tied to the particular coefficients in Theorem 5.6. For real $a, c \in [0, 1]$, direct expansion gives

$$\begin{aligned} ac^2 + \frac{24}{25}a^2 - \left(\frac{41}{25}a + \frac{4}{5}c - \frac{68}{75} \right) &= \frac{24}{25} \left(a - \frac{41 - 25c^2}{48} \right)^2 \\ &+ \frac{(1 - c)(5c - 3)^2(5c + 11)}{480}. \end{aligned} \quad (\text{B.8})$$

Both terms on the right are nonnegative. The equality atoms $(1/3, 1)$ and $(2/3, 3/5)$, each with probability $1/2$, have mean $(1/2, 4/5)$ and scalar envelope value $83/150$. For $25 \mid m$, use groups of m variables with these two means and $f_m = AE_2(W) + (24m/25)E_2(U)$. At binary normalized counts,

$$\frac{2f_m}{m^3} = ac^2 + \frac{24}{25}a^2 - \frac{ac + (24/25)a}{m}.$$

The expected correction is at most $49/(50m)$ because $ac \leq a$. The scaled concave envelope and termwise gap are, respectively, $(49/50)(1 - 1/m)$ and $(22/25)(1 - 1/m)$; cubic terms have lower envelope $1/10$, while quadratic terms have lower envelope zero. The affine minorant gives scaled hull gap at most $49/50 - 83/150 = 32/75$. Conditional independent sampling from the equality atoms, as in Theorem 5.6, gives scaled hull gap at least $(32/75)(1 - 1/m)$. Thus

$$\frac{33}{16}(1 - 1/m) \leq \frac{T(f_m)}{H(f_m)} \leq \frac{33}{16}.$$

At $m = 50$ the lower certificate is $1617/800 > 2$. This variant has a smaller limiting ratio than the two-level family in the main text.

C Exact finite cubic certificates

The following certificates use success counts, so their verification does not enumerate all 2^{192} binary vectors. Within each group, every binary vector with the same count has the same polynomial value. The finite enumeration therefore covers every vertex, including every possible dependence pattern relevant to the envelope.

For the six-term form (5.9), let $\Phi_m(A, B, C)$ be its binary count polynomial as defined in the proof of Proposition 5.5. Table 4 gives integers D, b_A, b_B, b_C, b_0 such that

$$D\Phi_m(A, B, C) \geq b_AA + b_BB + b_CC + b_0 \quad (0 \leq A, B, C \leq m \text{ integers}). \quad (\text{C.1})$$

Table 5 gives attaining laws. Each listed weight is nonnegative, the weights sum to one, and their mean count vector is $(m/4, m/2, 3m/4)$. Uniform success subsets conditional on these counts

m	D	b_A	b_B	b_C	b_0
6	13	962	778	842	-4816
8	7	793	770	798	-5882
64	105	871710	900446	899046	-51743768

Table 4: Integer affine minorants for the three-group witnesses.

m	Count states (A, B, C)	Corresponding probabilities
6	$(1, 4, 4), (2, 1, 6), (6, 2, 1)$	$17/26, 4/13, 1/26$
8	$(1, 2, 8), (1, 5, 6), (8, 4, 2)$	$2/7, 4/7, 1/7$
64	$(4, 19, 64), (5, 19, 64), (16, 40, 43), (64, 31, 17)$	$241/735, 12/735, 419/735, 63/735$

Table 5: Primal laws attaining the affine bounds.

therefore give the required singleton marginals. The checker below verifies the inequality at every integer count triple and equality of primal and dual expectations in rational arithmetic.

For $m = 8$, minimizing the residual in (C.1) over A, B for each $C = 0, \dots, 8$ gives

$$168, 42, 0, 30, 22, 0, 0, 16, 0.$$

The three primal states have values 405, 507, 734 respectively, with expected value 3572/7. In all three examples the orbit sizes, upper values per unweighted monomial, and lower values per unweighted monomial are respectively

$$s = \left(\binom{m}{3}, m \binom{m}{2}, \binom{m}{2}, m^2, m^2, \binom{m}{2} \right),$$

$$u = (3/4, 1/2, 1/2, 1/4, 1/4, 1/4), \quad l = (1/4, 0, 0, 0, 0, 0).$$

Thus $\text{cav } f = \sum_i c_i s_i u_i$ and $\sum_e \text{vex } f_e = \sum_i c_i s_i l_i$. These formulas prove the remaining columns of Table 2. The exact hull gaps for dimensions 18, 24, 192 are 10411/52, 3225/7, 27562048/105, respectively.

For the two-level $m = 16$ certificate (5.17), the integer residual is

$$2 \left[A \binom{C}{2} + 20 \binom{A}{2} \right] - 428A - 177C + 3372.$$

Its minimum over $C = 0, \dots, 16$ for successive $A = 0, \dots, 16$ is

$$540, 352, 204, 96, 28, 0, 9, 7, 0,$$

$$0, 19, 63, 132, 235, 369, 540, 742.$$

At counts $(8, 12)$ the value and its affine bound both equal 1088.

For the smaller members of (5.13), write $\Phi_m(A, C) = A \binom{C}{2} + (5m/4) \binom{A}{2}$. Table 6 gives affine minorants valid at every integer pair $0 \leq A, C \leq m$. At the listed counts each minorant equals Φ_m . Independently choosing uniform success subsets of those fixed sizes in the two groups gives every singleton mean $1/2, 3/4$ and attains the bound. Taking expectations of the minorant under any feasible law therefore proves the stated exact convex-envelope value. Every term has lower envelope zero and upper envelope $1/2$, so $\text{cav } f_m = T(f_m) = (9m/8) \binom{m}{2}$; subtraction gives the exact ratios. The checker verifies all $5^2 + 9^2 + 13^2$ minorant inequalities and these attainment and envelope equalities in rational arithmetic.

m	Affine minorant	Counts (A, C)	$\text{vex } f_m$	$\text{cav } f_m = T$	T/H
4	$9A + 4C - 19$	$(2, 3)$	11	27	$27/16$
8	$47A + 20C - 188$	$(4, 6)$	120	252	$21/11$
12	$(231/2)A + 48C - 684$	$(6, 9)$	441	891	$99/50$

Table 6: Exact smaller two-level certificates and attaining counts.

Here is a complete Python 3 checker using only the standard library. It enumerates $7^3 + 9^3 + 65^3$ three-group states and $5^2 + 9^2 + 13^2 + 17^2$ two-group states. The data are exactly those in the manuscript tables; no numerical optimization or proprietary solver is involved. The same executable text is supplied as `verification/check_stage02_finite.py`. Concatenate the two code blocks in order to obtain that file.

```

from fractions import Fraction as Q
from itertools import product
from math import comb

# m, coefficients, denominator, integer affine coefficients,
# probability denominator, (state, probability numerator), ratio
cases = [
    (6, (2,3,9,10,7,7), 13, (962,778,842,-4816), 26,
     [((1,4,4),17), ((2,1,6),8), ((6,2,1),1)],
     Q(20891,10411)),
    (8, (2,3,13,12,8,7), 7, (793,770,798,-5882), 7,
     [((1,2,8),2), ((1,5,6),4), ((8,4,2),1)],
     Q(6601,3225)),
    (64, (2,3,120,105,70,63), 105,
     (871710,900446,899046,-51743768), 735,
     [((4,19,64),241), ((5,19,64),12),
      ((16,40,43),419), ((64,31,17),63)],
     Q(7443345,3445256))]

def phi(a, b, c, coef):
    values = (comb(c,3), b*comb(c,2), comb(b,2),
              a*c, a*b, comb(a,2))
    return sum(v*w for v,w in zip(values,coef))

```

```

for m,coef,D,dual,P,atoms,ratio in cases:
    for a,b,c in product(range(m+1), repeat=3):
        rhs = sum(v*w for v,w in zip((a,b,c,1),dual))
        assert D*phi(a,b,c,coef) >= rhs
        assert all(p >= 0 for _,p in atoms)
        assert sum(p for _,p in atoms) == P
        means = [sum(Q(p,P)*state[j] for state,p in atoms)
                  for j in range(3)]
        assert means == [Q(m,4), Q(m,2), Q(3*m,4)]
        primal = sum(Q(p,P)*phi(*state,coef) for state,p in atoms)
        lower = sum(Q(v,D)*w for v,w in zip(dual,means+[1]))
        assert primal == lower
        sizes = (comb(m,3), m*comb(m,2), comb(m,2),
                  m*m, m*m, comb(m,2))
        upper = (Q(3,4), Q(1,2), Q(1,2), Q(1,4), Q(1,4), Q(1,4))
        cav = sum(c*s*u for c,s,u in zip(coef,sizes,upper))
        term_lower = Q(coef[0]*comb(m,3),4)
        assert cav > primal
        assert (cav-term_lower)/(cav-primal) == ratio
        assert ratio > 2
        print(3*m, cav, term_lower, primal, ratio)

# m, affine coefficients (A,C,1), attaining value, cav=T, ratio
small = [
    (4, (9,4,-19), 11, 27, Q(27,16)),
    (8, (47,20,-188), 120, 252, Q(21,11)),
    (12, (Q(231,2),48,-684), 441, 891, Q(99,50))]
for m,dual,vex,cav,ratio in small:
    def value(a,c):
        return a*comb(c,2)+Q(5*m,4)*comb(a,2)
    slacks = [value(a,c)-dual[0]*a-dual[1]*c-dual[2]
              for a,c in product(range(m+1), repeat=2)]
    assert min(slacks) == 0
    a,c = m//2,3*m//4
    assert Q(a,m) == Q(1,2) and Q(c,m) == Q(3,4)
    assert value(a,c) == dual[0]*a+dual[1]*c+dual[2] == vex
    assert (m+Q(5*m,4))*comb(m,2)/2 == cav
    assert cav > vex and Q(cav,cav-vex) == ratio < 2
    print(2*m, cav, vex, ratio)

mins = []
for a in range(17):
    residuals = [2*(a*comb(c,2)+20*comb(a,2))
                 -428*a-177*c+3372 for c in range(17)]
    mins.append(min(residuals))
assert mins == [540,352,204,96,28,0,9,7,0,0,19,63,
                132,235,369,540,742]
assert 8*comb(12,2)+20*comb(8,2) == 1088
assert 214*8+Q(177,2)*12-1686 == 1088
assert Q(2160,2160-1088) == Q(135,67)
assert Q(135,67) > 2
assert Q(243,115)*(1-Q(1,20)) == Q(4617,2300) > 2
print("All finite cubic certificates passed.")

```

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